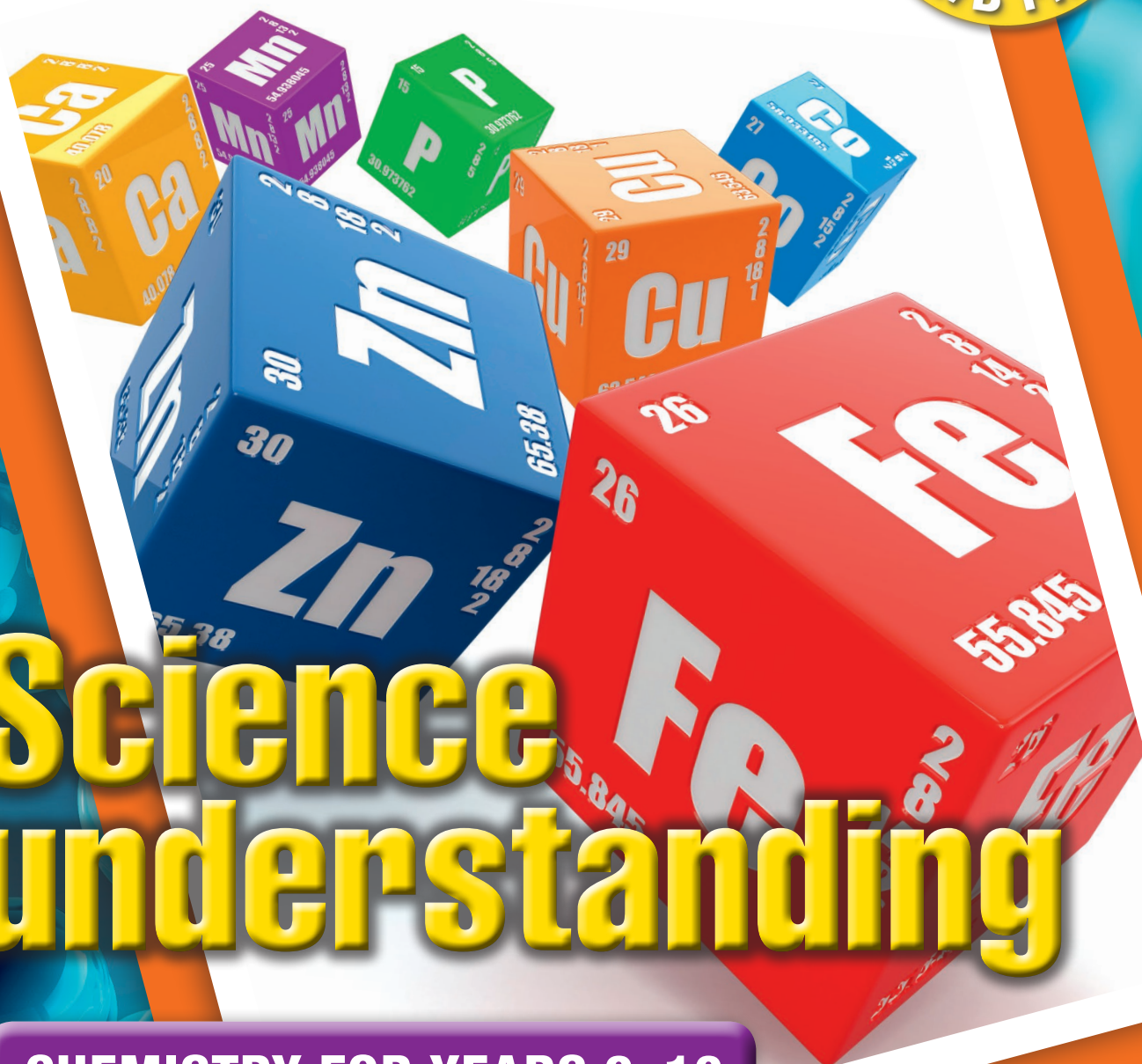


ACHIEVE!



Science understanding

CHEMISTRY FOR YEARS 9–10

Janine Haeusler



Made for the
Australian Curriculum





ACHIEVE!

Science understanding

CHEMISTRY FOR YEARS 9-10

Janine Haeusler

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The *Achieve!* series of science titles has been developed for teachers who require modified, curriculum-focused activities for secondary school students who struggle to read and comprehend regular classroom materials because of poor literacy skills. Many of the pages in *Chemistry for years 9–10* will assist teachers to develop lessons for the chemistry sub-strand in *Science understanding* for years 9 and 10. Units of work cover a variety of topics including atomic structure, the periodic table, compounds, acids and bases, rates of reactions and conservation of mass. Students build their scientific vocabulary and understanding of key concepts. Work sheets involving practical activities also develop a student's ability to plan and safely conduct experiments, and to record and analyse data. Practical activities are clearly marked with the work safely symbol ▲.

The CD-ROM contains all the pages that are featured in the book, as well as a file of illustrations that can be used for other work sheets. These documents can be easily edited to suit the individual needs of students.

The teacher's notes provide valuable guidance about using each unit, and contain the following elements.

Objectives – These are the key skills and knowledge learnt through using the work sheets.

Prior knowledge – This refers to the skills and knowledge required for students to complete the tasks. Some activities are more challenging than others, however, as a rule, students should have a reading comprehension age of six to nine years to be able to attempt the activities.

Background – This section provides additional guidance to help teachers to present a successful lesson. It may include definitions or further background notes about the subject matter.

Starter activity – Warm-up activities are provided to put the content into proper context and to gain student interest and attention.

Resource and Activity work sheets – The Resource sheets do not always contain activities and are used either to stimulate discussion or as a part of the unit's activities. The Activity sheets may be used sequentially or as stand-alone tasks, depending on the needs of the students.

Extension ideas – Additional tasks are suggested to recap on the main skills or knowledge taught in the lesson, as well as ways to extend, assess or include technology in the lesson.

The Australian Curriculum alignment	Unit									
	1	2	3	4	5	6	7	8	9	10
Year 9										
All matter is made of atoms which are composed of protons, neutrons and electrons; natural radioactivity arises from the decay of nuclei in atoms (ACSSU177)	✓									
Chemical reactions involve rearranging atoms to form new substances; during a chemical reaction mass is not created or destroyed (ACSSU178)			✓	✓		✓	✓		✓	✓
Chemical reactions, including combustion and the reactions of acids, are important in both nonliving and living systems and involve energy transfer (ACSSU179)					✓	✓	✓			
Year 10										
The atomic structure and properties of elements are used to organise them in the periodic table (ACSSU186)	✓	✓	✓							
Different types of chemical reactions are used to produce a range of products and can occur at different rates (ACSSU187)						✓	✓	✓	✓	✓

Objectives

- Describe and model the structure of atoms in terms of the nucleus, protons, neutrons and electrons
- Compare the mass and charge of protons, neutrons and electrons
- Describe the structure of atoms in terms of electron shells

Prior knowledge

Students should be aware that all matter is made up of particles called atoms. They should also understand that a material in which all the atoms are the same is called an element, and that elements are listed on a chart called the periodic table. Students should know the symbols for common elements.

Background

Atoms are the smallest units of matter. Each atom has a central nucleus with electrons orbiting around it. The nucleus, which is made up of protons and neutrons, gives the atom most of its mass. The atomic number and atomic mass of an element can be found on the periodic table. The atomic number represents the number of protons an element has. The atomic mass is the total number of neutrons and protons in the nucleus of its atom. The orbiting electrons are found in shells. There are a number of shells around the nucleus and each shell can hold a certain number of electrons.

Starter activities

Introduce atoms with an introductory DVD/video from your school's library, or watch a video available online, such as the following YouTube clip, up to the 4.30-minute mark: <http://www.youtube.com/watch?v=h6LPAwAmnCQ>

Resource and Activity sheets

The Resource sheet 'An introduction to atomic structure' and the Activity sheet 'Atomic structure' develop students' understanding of atomic structure. The Resource sheet explains the sub-atomic particles that make up an atom and how these sub-atomic particles are arranged. Students use this information to help them complete a cloze activity and a matching phrases task on the Activity sheet.

The Resource sheet 'Atomic structure and the periodic table (1)' and the Activity sheet 'Atomic structure and the periodic table (2)' aim to familiarise students with the information provided by the periodic table. Students are then asked to identify the atomic number and atomic mass for different elements and calculate the number of protons, electrons and neutrons in the atoms of various elements. Refer to the periodic table on the inside front cover when completing these work sheets.

The Activity sheet 'Arrangement of electrons' describes how electrons are arranged in shells around the outside of the nucleus. Students are required to label a diagram of a lithium atom and answer some short-answer questions. Students then complete a table and draw the arrangement of electrons for an aluminium atom in their workbook.

Extension ideas

- Students could build a model of an atom of a particular element.
- Students could compose their own song about atoms and elements. There are numerous examples available online that they could refer to.
- Students could make edible models of various atoms using Smarties and M&Ms.

Selected answers

Atomic structure 3 1, 0, 1, no neutrons.

Atomic structure and the periodic table (2):

1 hydrogen: H, 1, 1, 1, 1, 0 carbon: C, 6, 12, 6, 6, 6 oxygen: O, 8, 16, 8, 8, 8 **2** 28, 28, 59, nickel

Arrangement of electrons 2 2, 8, 3



Work safely

Teachers should conduct their own risk assessments for any activity or experiment they choose to do with their classes. All experiments should be carried out under close supervision.

Suggested web addresses were functional at the time of publication but teachers should check them for inappropriate content and to ensure that they are still functional before use with students.

An introduction to atomic structure

All matter is built from particles called **atoms**. Atoms are the building blocks of all materials on earth. More than 100 types of atoms have been discovered. We call them **elements**.

The structure of an atom

For a long time people thought that atoms were the smallest particles in existence. We now know that atoms are made up of even smaller **sub-atomic particles**: electrons, protons and neutrons.

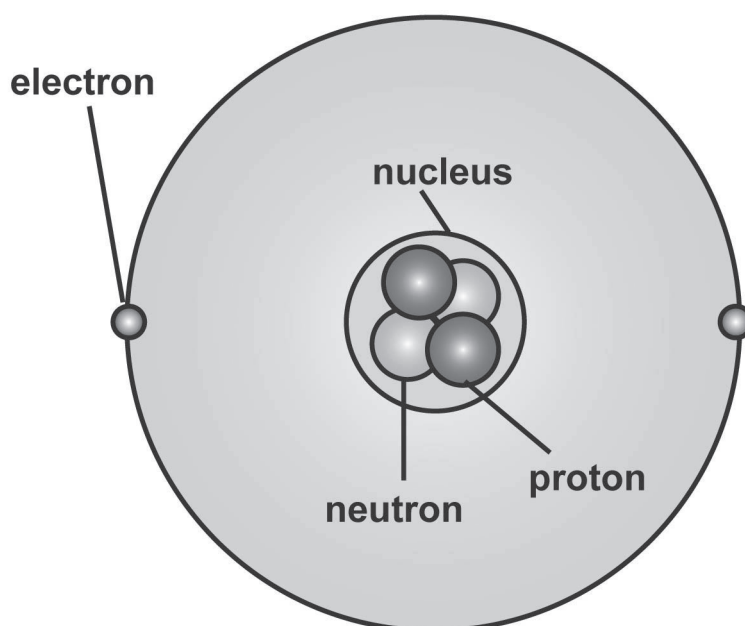
- **Electrons** are very tiny and light. They have a **negative** charge.
- **Protons** are much larger and heavier than electrons and have a **positive** charge.
- **Neutrons** are larger and heavier like protons, but have no electrical charge (they are **neutral**).

How are the particles arranged?

An atom has a **nucleus** in its centre. Protons and neutrons can be found inside the nucleus. Electrons are constantly spinning around the outside of the nucleus. Each atom has an equal number of protons and electrons. This means it has no overall **charge**.

Scientists use simple diagrams to show the structure of an atom. This is a diagram of a helium atom. It contains:

- 2 protons
- 2 neutrons
- 2 electrons.



Atomic structure

Read the Resource sheet 'An introduction to atomic structure', then answer the following questions.

 **1** Use the words in the word bank to complete the following passage.

Some words will be used more than once.

Elements are simple s_____ made up of the same type of a_____. There are more than 1_____ different types of elements. Each element's atom is u_____. For example, c_____ atoms are different from silver atoms.

Atoms are made up of t_____ different sub-atomic particles: n_____, protons and e_____. The middle part of the atom is called the n_____. The nucleus contains p_____ and neutrons. The electrons are found s_____ around the outside of the n_____.

An atom has the same number of protons and e_____. For example, hydrogen has one proton, so it also has o_____ electron.

Word bank

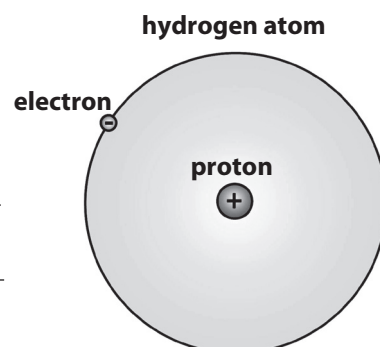
three
atom
electrons
nucleus
substances
100
neutrons
spinning
one
copper
protons
unique

2 Match the sentence halves to complete the information below.

Protons and neutrons are	no charge. This means they are neutral.
Protons have a	can be found spinning around the nucleus.
Electrons are tiny and	positive charge.
Neutrons have	about the same size and mass.
Electrons have	a negative charge.

3 Look at the diagram of a hydrogen atom.

- How many protons does a hydrogen atom have? _____
- How many neutrons does a hydrogen atom have? _____
- How many electrons does a hydrogen atom have? _____
- What is unusual about hydrogen? _____



Atomic structure and the periodic table (1)

The **periodic table** is a record of all the **elements**. The periodic table gives us lots of information about each element. It tells us:

- each element's symbol
- the number of **protons** in an element's atom. This is called its **atomic number**. The atomic numbers increase by one as you go along the periodic table.
- the **atomic mass** of an element. This tells us the mass of an element compared to carbon. It is approximately the total number of protons and **neutrons** added together.
- the number of **electrons** in an element's atom. The atomic number tells us this, too, because this number is always the same as the number of protons.
- some of the properties of different elements.

6	Atomic number
C	Symbol
Carbon	Name
12.01	Atomic mass

From looking at the information provided for carbon, we know that carbon:

- has the symbol C
- has an atomic number of 6. This means it has 6 protons and 6 electrons.
- has an atomic mass of 12. This means it has 6 neutrons (atomic mass minus the number of protons).

Atomic structure and the periodic table (2)

Read the Resource sheet 'Atomic structure and the periodic table (1)' and then answer the following questions.

1 H Hydrogen 1.01	6 C Carbon 12.01	8 O Oxygen 16.00
--	---	---



- 1 The diagram above is from the periodic table. Use the information provided to complete the table below. Round to the nearest whole number.

	Hydrogen	Carbon	Oxygen
Symbol			
Atomic number			
Atomic mass			
Number of protons			
Number of electrons			
Number of neutrons			

- 2 An atom contains 28 protons and 31 neutrons.

a What is its atomic number? _____

b How many electrons does it have? _____

c What is the atomic mass of the atom? _____

d Use the periodic table on the inside front cover to find the name of the element. _____

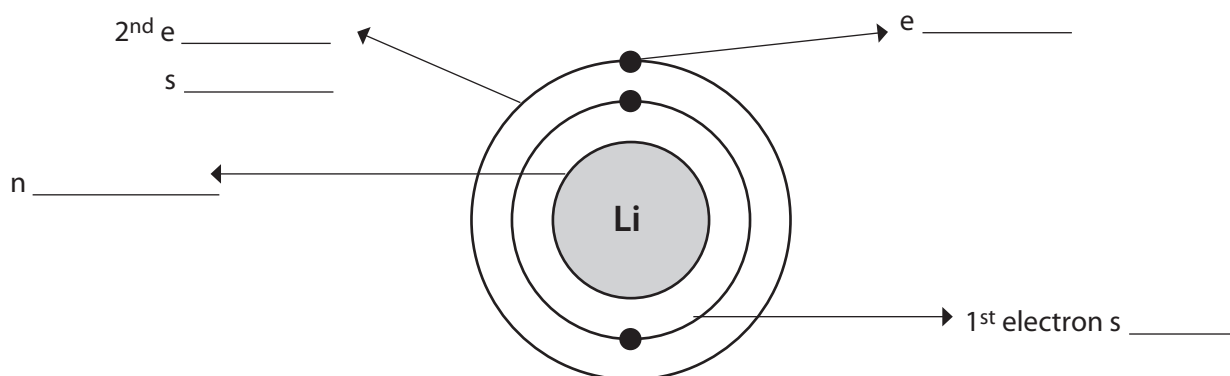
Arrangement of electrons

Electrons are always moving. They spin quickly around the outside of the **nucleus** in rings, or **shells**.

Not all shells can hold the same number of electrons.

- 1st ring (closest to the nucleus): maximum of 2 electrons
- 2nd ring: maximum of 8 electrons
- 3rd ring: maximum of 18 electrons
- 4th ring: maximum of 32 electrons

 **1** The diagram below shows how electrons are arranged in a lithium atom.



- What is the symbol for lithium? _____
 - Label the diagram.
 - How many electrons does lithium have? (Hint: Count the number of electrons in the diagram.) _____
 - How many electron shells does lithium have? _____
 - How many electrons are in the first shell? _____
 - How many electrons are in the second shell? _____
 - How many protons does lithium have in its nucleus? _____
- 2** If the element aluminium contains 13 electrons, how many are in each shell? Draw the electron arrangement for an aluminium atom in your workbook.

Objectives

- To understand that the elements in the periodic table are organised according to their atomic structure and properties
- To recognise that elements in the same group of the periodic table have similar properties
- To describe simple patterns in the arrangement of elements in the periodic table and relate these to electron structure
- To identify the different groups that elements are classified into

Prior knowledge

Students need to have a basic understanding of the periodic table. They should be aware that the numbers on the periodic table represent each element's atomic number (number of protons) and atomic mass (combined number of protons and neutrons). They should also understand that the number of protons in the atom of an element is equal to the number of electrons.

Background

The periodic table is a chart of the elements arranged by atomic number. Elements with similar properties are arranged in columns called groups, such as alkali metals, halogens and noble gases. The group number represents the number of electrons in the outer electron shell. This number affects the physical and chemical properties of an element. The horizontal rows are called periods. Elements in the same period have the same number of electron shells.

Starter activity

Play a game of 20 questions, having students use a process of elimination to work out which element on the periodic table you are thinking of. (Provide students with the periodic table on the inside front cover to refer to.)

Resource and Activity sheets

The Resource sheet 'How is the periodic table arranged?' illustrates the way the periodic table is arranged into periods and groups, and explains how this relates to the arrangement of electrons in elements. Students should refer to this page when

completing the Activity sheet 'Organising the periodic table.' The Activity sheet has some short-answer questions to complete, along with some true/false statements and a table to fill out.

After reading the Resource sheet 'Naming groups in the periodic table' and completing the Activity sheet 'Groups in the periodic table', students should be able to identify some of the different groups, including alkali and transition metals, semi-metals, halogens and noble gases. Students should be able to describe the characteristics of each group and classify elements into the different groups.

The Activity sheet 'Noble gases' explores a special group of elements called noble gases. Students complete a table exploring the arrangement of electrons in noble gases, then complete a number of higher-order questions, which help them to understand that noble gases are unreactive because their outer electron shells are full.

Extension ideas

- Students could complete an experiment exploring the reactivity of the Group 1 and 2 metals.
- Students could research metalloids and prepare a poster or brochure explaining how they are used in the electronics industry as semi-conductors.
- Students could devise a game of 'Bingo' using words to do with the periodic table.

Selected answers

Organising the periodic table 7 magnesium 12, 3, 2; iodine 53, 5, 7; aluminium 13, 3, 3

Noble gases 1 neon 2, 8; argon 2, 8, 8; krypton 2, 8, 8, 18; **2** Noble gases have a full outer shell, so are stable and unreactive. They do not need to fill their outer shell. **3** Sodium gives away one electron. **4** Chlorine gains, or receives, one electron.



Work safely

Teachers should conduct their own risk assessments for any activity or experiment they choose to do with their classes. All experiments should be carried out under close supervision.

How is the periodic table arranged?

The **periodic table** is arranged like a grid. The elements appear in order of increasing **atomic mass** and are placed in a specific location because of the way they look and act.

	Group 1 2																3	4	5	6	7	8 (or 0)		
Period	1																	2						
1	1 H																	2 He						
2	3 Li	4 Be															5 B	6 C	7 N	8 O	9 F	10 Ne		
3	11 Na	12 Mg															13 Al	14 Si	15 P	16 S	17 Cl	18 Ar		
4	19 K	20 Ca	21 Sc	22 Ti	23 V	24 Cr	25 Mn	26 Fe	27 Co	28 Ni	29 Cu	30 Zn	31 Ga	32 Ge	33 As	34 Se	35 Br	36 Kr						
5	37 Rb	38 Sr	39 Y	40 Zr	41 Nb	42 Mo	43 Tc	44 Ru	45 Rh	46 Pd	47 Ag	48 Cd	49 In	50 Sn	51 Sb	52 Te	53 I	54 Xe						
6	55 Cs	56 Ba		72 Hf	73 Ta	74 W	75 Re	76 Os	77 Ir	78 Pt	79 Au	80 Hg	81 Tl	82 Pb	83 Bi	84 Po	85 At	86 Rn						
7	87 Fr	88 Ra		104 Rf	105 Db	106 Sg	107 Bh	108 Hs	109 Mt	110 Ds	111 Rg	112 Uub	113 Uut	114 Uuq	115 Uup	116 Uuh	117 Uus	118 Uuo						
																			57 La 58 Ce 59 Pr 60 Nd 61 Pm 62 Sm 63 Eu 64 Gd 65 Tb 66 Dy 67 Ho 68 Er 69 Tm 70 Yb 71 Lu					
																			89 Ac 90 Th 91 Pa 92 U 93 Np 94 Pu 95 Am 96 Cm 97 Bk 98 Cf 99 Es 100 Fm 101 Md 102 No 103 Lr					

Rows

- The elements are placed in rows. All horizontal rows are called **periods**.
- All elements in the same period have the same number of **electron shells**.
- There are seven periods, but elements 57–71 and 89–103 have been placed underneath to make the table a more manageable shape.

Columns

- Elements are arranged in vertical columns called **groups** (1–8), with the **transition metals** between groups 2 and 3. For example, group 1 consists of H, Li, Na, K, Rb, Cs, Fr.
- The transition metals in the centre section behave a bit differently, so are not included.
- All elements except hydrogen in group 1 share similar properties. Hydrogen is a special element, so is often placed above group 1.
- The group number also represents the number of **electrons** in the **outer shell**. This number affects how **reactive** the element is. The most reactive elements have very few electrons in their outer shell, or only need a few more to fill it.

Organising the periodic table

Read the Resource sheet 'How is the periodic table arranged?', then answer the questions.

- 1 What do we call the columns? _____
- 2 How many periods are there on the periodic table? _____
- 3 Which group does Cl (chlorine) belong to? _____
- 4 In which period is Cu (copper) found? _____
- 5 Circle the correct ending for each of the sentences below.

The horizontal rows are called	groups.	periods.
The groups run	from left to right.	from top to bottom
All elements in the same period	have the same number of electrons.	have the same number of electron shells.
All elements in the same group	have the same number of electrons in their outer shell.	have the same number of electron shells.

- 6 Decide whether each of these statements is true or false.
 - a The element C (carbon) is found in period 3. _____
 - b The element K (potassium) is found in group 1. _____
 - c Na (sodium), P (phosphorus) and Cl (chlorine) atoms all have 4 electron shells. _____
 - d Mg (magnesium), Ca (calcium) and Ba (barium) all have 1 electron in their outer shell. _____
- 7 Complete the table below.

Element	Number of electrons	Number of electron shells	Number of electrons in outer shell
Mg (magnesium)			
I (iodine)			
Al (aluminium)			

Naming groups in the periodic table

Elements on the periodic table can be divided into metals and non-metals. They can be further broken down into different groups based on their properties.

The metals found in Group 1 are very **reactive**. This means they react violently with water. They are also soft and can be cut with a knife. Examples include lithium, sodium and potassium, which are also known as the **alkali metals**. Hydrogen is in Group 1 of the periodic table but it is not considered part of the alkali metals because it is a **non-metal**.

The metals found between Groups 2 and 3 are called **transition metals**. They have the typical properties of metals – they are hard, dense and have high melting and boiling points. The **semi-metals** or **metalloids** are located along the stepped line between metals and non-metals. Boron, silicon, germanium, arsenic, antimony, tellurium and polonium are all semi-metals. They are called semi-metals because they have some characteristics of metals and some of non-metals. For example, silicon is shiny, but it is brittle, so it would break if you hit it with a hammer.

The non-metals in Group 7 are very **reactive**. They are called **halogens**. Halogens are poisonous non-metals. Examples include fluorine, chlorine, bromine, iodine and astatine. Like most non-metals, they have low melting and boiling points.

Group 8 – sometimes known as Group 0 – is found on the right-hand side of the periodic table. These elements are very special because they do not react to form compounds. They are found by themselves in nature. They are called **noble gases**.

Groups in the periodic table

Use the Resource sheets 'How is the periodic table arranged?' and 'Naming groups in the periodic table' to answer the following questions.

 **1** On the periodic table on the 'How is the periodic table arranged?' Resource sheet, colour:

- B (boron), Si (silicon), Ge (germanium), As (arsenic), Sb (antimony), Te (tellurium) and Po (polonium) in green. These elements are called semi-metals, or metalloids.
- the alkali metals in yellow
- the transition metals in orange
- the halogens in blue
- the noble gases in pink
- all the remaining non-metals in purple. Remember, hydrogen is a non-metal.

2 Name the elements in Group 1. _____

3 What is unusual about the Group 1 elements? _____

4 Which group is Br (bromine) in? What do we call this group? _____

5 The metals between Groups 2 and 3 are called transition metals. How many transition metals are there? _____

6 Which of these elements is the odd one out? Circle it. Li, Na, Ca, K, Rb, Cs

7 Match the sentence halves to summarise the different groups.

Alkali metals react

Group 7 elements, also known as

Noble gases, found in Group 8 (or 0),

Transition metals have high melting

do not react to form compounds.

violently in water.

and boiling points and are dense.

halogens, are very reactive poisonous non-metals.

Noble gases

The **noble gases** are a special group of elements that exist as single atoms. They are very **stable** and rarely react with other elements. Electron shells each hold a different number of electrons. The noble gases have a full outer shell.

-  **1** Use the information below to help fill in the table. Keep in mind that a shell does not need to be completely filled before electrons begin to fill the next level.

Electron shells

1	2	3	4
holds 2 electrons	holds 8 electrons	holds 8 or 18 electrons	holds 18 or 32 electrons

		Number of electrons in each shell				
Element	Atomic number*	1	2	3	4	5
helium	2	2	-	-	-	-
neon	10					
argon	18					
krypton	36					

*Atomic number = number of electrons

When elements form a **compound** they fill their outer shell to become stable. Elements can react with each other to form compounds by giving, receiving or sharing electrons.

- 2** Can you explain why noble gases do not form compounds? _____
- _____
- 3** Sodium has one electron in its outer shell. What do you think sodium does with this electron when it forms a compound? _____
- 4** Chlorine has 7 electrons in its outer shell. What do you think chlorine needs to do to fill its outer shell when it forms a compound? _____

Objectives

- To understand that different metals have different properties and that we use different metals in different ways according to their properties
- To realise that the properties of metals can be improved by combining them with other metals and non-metals to form alloys
- To understand that metals are found in the ground and need to be extracted by mining
- To understand that the natural compounds of metals are called ores

Prior knowledge

Students should have a basic understanding of some of the general characteristics of metals, such as high melting and boiling points, shiny, hard and dense.

Background

A metal's properties determine the qualities of the metal. Metals have different properties and so have different uses. Different metals can be combined in mixtures to form alloys. Alloys have different properties from the individual metals that comprise them, making them useful for different jobs. Some metals are more reactive than others. For example, gold is unreactive, while sodium is very reactive. Unreactive metals may be found in earth's crust by themselves. Reactive metals exist as compounds called ores. Mining is used to extract these ores.

Starter activity

Brainstorm some of the general characteristics of metals. Ask students to name some unusual metals and to explain why they are unusual. Why is mercury different from other metals? What is different about sodium and potassium?

Resource and Activity sheets

In the Activity sheet 'Properties of metals', students read a passage and underline the different characteristics of metals. They then match each characteristic to its definition. Students complete higher-order questions that require students to relate the properties of metals to their uses.

The Activity sheet 'Properties of different metals' presents an incomplete table listing the properties and uses of different metals. Students need to fill in the missing vowels. They then answer questions in a game of 'What metal am I?'

The 'Alloys' Activity sheet explains how metals can be combined to form alloys. Students identify spelling errors in a passage about alloys, then unjumble sentences.

The Activity sheet 'Mining' explains how metals must be removed from the ground by the process of mining and then extracted from their ores. Students are given a passage and must choose the correctly spelled word in each pair. They are then provided with a number of pieces of information and must decide whether the information refers to open-cut or underground mining.

In the 'Corrosion' Activity sheet, students can read a passage on the topic of corrosion and then complete activities exploring different ways of preventing corrosion.

Extension ideas

- In groups, have students play 20 questions, using a process of elimination to identify the metal or alloy.
- Students could watch a DVD or complete a research assignment about mining in Australia.
- Students could conduct an experiment on preventing corrosion, or identifying the causes of corrosion.



Work safely

Teachers should conduct their own risk assessments for any activity or experiment they choose to do with their classes. All experiments should be carried out under close supervision.

Properties of metals

 **1** Read the information below and underline the different **properties** of metals.

Metals are good conductors of heat and electricity. They also have high melting and boiling points, and high density. Most metals are ductile, which means they can be drawn into wires. Metals have lustre: when they are cut or polished they are shiny. They are very useful because as well as being strong, they are malleable – most metals can be hammered into sheets or heated and then shaped. Metals have good tensile strength, too – they cannot be easily stretched or pulled apart. All metals are sonorous; they make a ringing sound when hit.

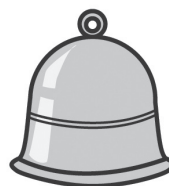
2 Draw a line to match each property with its definition.

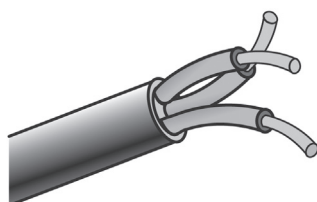
Property	Definition
dense	Makes a ringing sound when hit
ductile	Can be hammered into shape
tensile strength	Heavy for its size
malleable	Cannot be easily stretched or pulled apart
sonorous	Can be drawn into wires

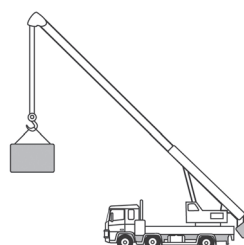
Metals are good conductors of heat and electricity. This means that they allow heat and electricity to move through them easily.

3 The pictures below show some of the uses of metals. Write one property that makes metal useful for each object.









Properties of different metals

Different metals have different properties that make them useful for different jobs.

 **1** Fill in the missing vowels (a, e, i, o, u) to complete the information in the table.

Metal	Properties	Uses
al _ m _ n _ _ m	s _ ft, l _ ght, resists c _ rr _ si _ n, good el _ ctr _ c _ l conductor	s _ ft dr _ nk cans, w _ nd _ w frames, _ _ rcraft
c _ pp _ r	g _ _ d cond _ ct _ r of h _ _ t and electricity, d _ ct _ le, r _ sts slowly	w _ t _ r pipes and t _ nks, electr _ c _ l w _ ring
g _ ld	s _ ft, d _ nse, m _ ll _ _ ble, resists corrosion, has l _ str _	j _ w _ ll _ ry
_ r _ n	hard, br _ ttle, r _ sts easily, m _ gn _ tic	used to make st _ _ l, br _ dg _ s

2 What metal am I?

- a** I am a light metal which does not corrode easily. I can be found in aeroplanes and window frames. I am _____.
- b** I am a hard metal often used in buildings. I rust easily, so you can often find me mixed with other metals and non-metals to slow down rusting. I am _____.
- c** I am a shiny and soft metal. I am very valuable and used for jewellery and crowns. I am _____.
- d** I am an excellent conductor of electricity, so I am often used in electrical wiring. You will often find hot water tanks made out of me because I resist corrosion. I am _____.

Alloys

-  **1** Ten of the words in the passage below have been spelt incorrectly. Correct them and then re-read the text.

(Metils) _____ can be changed by adding other metals, and sometimes even non-metals, to them. Some of the metals that we use are (pur) _____ but many are a (mixtir) _____ of different metals. A mixture of two or more metals is called an (aloy) _____. Often the metals are mixed together to make use of their different (propertes) _____. The resulting **alloy** has different properties to the metals it (contains) _____ when used on their own. Solder is an alloy that is used to join metals that require good (electrial) _____ conductivity. It contains tin and lead. Solder has a lower (mealting) _____ point than tin and lead and its (hardeness) _____ is also (differant) _____ from tin or lead.

- 2** Unjumble the sentence parts and then rewrite the information about alloys in the correct order.

two or more a mixture of An alloy is metals.

the individual metals used to make them. different properties than Alloys have

contains the metals Solder tin and lead.

non-metals contain Some alloys as carbon. such

Mining



1 Circle the correctly spelled word in each pair.

Only a few metals are found in a pure form in nature/nature. Most of the metals we use come from minerals/minerals or **ores** found in the ground. Most ores are **oxides** of metals. That means that the metal is combined with oxygen/oxygen. Once these ores have been mined/mined, the metals need to be extracted/extracted from them. The method of mining used depends on the size, shape and depth/depth of the ore body.

2 The two methods of mining are open-cut (O) and underground (U). Decide which method each sentence describes and label each one.

_____ Used when ore is close to the surface

_____ Big hole in the ground

_____ Used when ore body lies deep beneath the earth

_____ Used when ore body is narrow and at a steep angle

_____ Leaves less scars on the surface of earth

_____ Dug in steps or levels to reach the area to be mined

_____ More cost effective

_____ Tunnels or shafts are built

_____ Steps include roads. As more material is removed, new steps are built

_____ More dangerous and expensive than open-cut mining

3 Complete the table below about metals commonly mined in Australia.

Metal	Name of ore	Formula	Elements present
	bauxite	Al_2O_3	1. Aluminium 2.
	haematite	Fe_2O_3	1. 2. Oxygen
	galena	PbS	1. 2.
	gold	Au	1.

Corrosion

Metals are some of the most important materials used on earth. One problem with metals is that they can be attacked by water and oxygen. This problem is called **corrosion**. When corrosion occurs, small holes appear on the surface of the metal and the strength of the metal decreases.

Iron corrodes more quickly than most other metals. Another name for the corrosion of iron is rust. Corrosion is a chemical reaction, which can be written in a word equation.



When some metals, such as aluminium and nickel, corrode, the oxides form a protective coat around the metal. When iron rusts, oxygen can keep attacking the iron because the iron oxide forms flakes of rust, which fall off. Eventually, all the iron will be attacked.

-  **1** The table below lists ways that rust can be prevented. Match each method to its description.

Method	Description
painting	Covering iron with a thin layer of another metal, such as tin or chromium, can stop rusting. When a metal is covered with zinc it is called galvanising.
oiling	When a more reactive metal is connected to iron, the reactive metal corrodes instead of the iron. The iron is protected at the expense of the more reactive metal.
plating	A thin layer of paint stops oxygen and water reaching the metal.
alloying	A thin layer of plastic stops oxygen and water touching the metal.
plastic coating	A layer of oil stops oxygen and water attacking the iron.
sacrificial protection	Mixing iron with other metals, such as chromium, protects it from rusting.

- 2** Decide which method above should be used to protect each of the following objects from corrosion.

- | | | | |
|-----------------------------|-------|----------------------|-------|
| a bike chain | _____ | d gate | _____ |
| b tin of baked beans | _____ | e shed roof | _____ |
| c car bumper | _____ | f metal knife | _____ |

Objectives

- To understand that a compound contains atoms of different elements combined together
- To realise that the different elements in a compound exist in fixed amounts
- To identify the different elements in compounds and name different chemical compounds when provided with the chemical formulas
- To be able to name common radicals
- To recognise chemical formulas and draw molecular diagrams

Prior knowledge

Students should know the symbols for common elements and be able to give the formulas for common compounds such as water and table salt.

Background

Compounds contain atoms of different elements combined together in fixed amounts. To represent a compound, we write the symbols of its elements together in a formula. The formula is made up of the chemical symbols of the elements present and shows the type and number of atoms found in one molecule of the compound. When naming a compound, the metal is named first and the non-metal is given the ending 'ide'. If oxygen is present the name often ends with 'ate' instead.

Starter activity

Introduce the topic of compounds by asking students, 'If there are only about 100 types of elements in the world, why are there so many different kinds of substances?' What do we call substances that are made up of different elements combined together? Ask students to name some compounds, then ask if there is another way of saying them, e.g. water: H_2O . Write any names and formulas on the board. What do the formulas mean and what can they tell us about a compound? Have students complete a Venn diagram comparing elements to compounds.

Activity and Resource sheets

In the Activity sheet 'Molecules and compounds' students learn the difference between molecules of elements and molecules of compounds and practise

identifying the atoms present in different compounds. Students use the word bank to complete the passage and identify elements present in different compounds.

The Resource sheet 'Naming compounds (1)' explains the different rules students must follow to name compounds when provided with chemical formulas. Students should refer to this page when completing the Activity sheet 'Naming compounds (2)'. The Activity sheet has a small cloze exercise and a table of chemical formulas, which students must name.

The Activity sheet 'Radicals' familiarises students with common radicals. Students complete the table provided and then circle the correct formula for different compounds.

After completing the Activity sheet 'Naming molecules', students should be able to match names of compounds to the correct formulas and molecular diagrams.

Extension ideas

- Students could build an atom using simulations found on the internet.
- Students could build molecules of different compounds using molecular kits.

Selected answers

Radicals: 2 MgCO_3 , ZnSO_4 , NH_4Cl , KNO_3 , NaHCO_3 , NaOH

Molecules and compounds

 **1** Use the words in the word bank to complete the following passage.

All matter is made up of a _____. Atoms are rarely found on their own. They are often found joined together, forming m_____.

In molecules, the atoms that are chemically joined together may all be the same or they may be d_____. We can have molecules of an e_____ – for example, hydrogen (H_2) or oxygen (O_2) – or we can have molecules of a c_____. Compounds are substances that are made up of t_____ or more elements chemically joined together. For example, w_____ (H_2O) and ammonia (NH_3) are both compounds. The s_____ H_2O and NH_3 are the formulas for water and ammonia. A chemical formula tells us which t_____ of atoms are present and how m_____ atoms of each type there are per molecule.

Word bank

symbols
atoms
different
molecules
two
water
type
element
many
compound

2 Classify each of the following as either a molecule of an element or a molecule of a compound.

P_4	_____	I_2	_____
HCl	_____	CO	_____
CH_4	_____	S_8	_____

3 Complete the following information. The first one has been done for you.

Substance	Formula	Each molecule contains
water	H_2O	2 atoms of hydrogen, 1 atom of oxygen
magnesium oxide	MgO	
salt (sodium chloride)	$NaCl$	
carbon dioxide	CO_2	
sugar	$C_{12}H_{22}O_{11}$	
oxygen gas	O_2	
ammonia gas	NH_3	

Naming compounds (1)

Most chemical **formulas** for **compounds** have numbers in them, as well as the symbols for different elements. These numbers tell us how many atoms of each element there are in one molecule of the compound.

Rule 1

When two elements are present:

- the metal is named first
- the elements will combine to make a compound ending in 'ide'
- if both are non-metals and one of them is hydrogen, hydrogen is written first.

For example:

KBr	potassium brom ide
MgO	magnesium ox ide
NaCl	sodium chlor ide
CaS	calcium sulf ide

Rule 2

When two or more elements combine with oxygen:

- the metal is named first
- the compound often ends with part of the name of the second element, combined with 'ate'.

For example:

Na ₂ CO ₃	sodium carbon ate
ZnSO ₄	zinc sulf ate
NaNO ₃	sodium nitr ate
MgSO ₄	magnesium sulf ate

Rule 3

When there are only two elements and both are non-metals:

- if hydrogen is present, it is written first
- prefixes are joined to the name of the second element to show how many atoms of it there are. The prefixes are:
 - di – 2 • tri – 3 • tetra – 4 • penta – 5.

Mono means 1. The '1' is commonly left out, because scientists assume that no prefix means that there is only one of that atom.)

For example:

CO ₂	carbon d ioxide	CO	carbon mon oxide
SO ₃	sulfur tri oxide	HBr	hydrogen bromide

Naming compounds (2)

Use the information on the Resource sheet 'Naming compounds (1)' to answer the following questions.

 **1** Fill in the missing words in the sentences below.

- a** When two e_____ combine, the name usually ends in 'i_____':
- b** When two elements combine, the m_____ is written first.
- c** When two or more elements combine with o_____, the name of the compound usually ends in 'a_____':
- d** If two elements are present and both are n_____ -metals, we use a p_____ on the second element.
- e** 'Mono' means o_____.

2 Name the compound that magnesium will make with each of the following elements:

- a** chlorine _____
- b** oxygen _____
- c** fluorine _____

3 Complete the table below.

Rule number	Chemical formula	Name of compound
Rule 1	NaBr	
	KI	
	Na ₂ O	
Rule 2	Na ₂ CO ₃	
	H ₂ SO ₄	
	CuSO ₄	
Rule 3	SO ₂	
	CO	
	NO ₃	

Radicals

Many chemical **compounds** contain a chemical **radical**. A chemical radical is a group of atoms that tend to stay together when the compound reacts with another compound. The names of many chemical compounds include the name of the radical they contain.



1 Complete the table below. The first one has been done for you.

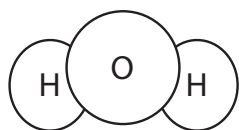
Name of common radical	Formula of radical	Contains
carbonate	CO ₃	1 atom of carbon, 3 atoms of oxygen
hydrogen carbonate	HCO ₃	
hydrogen sulfate	HSO ₄	
hydroxide	OH	
nitrate	NO ₃	
sulfite	SO ₃	
sulfate	SO ₄	
ammonium	NH ₄	
phosphate	PO ₄	

2 Circle the correct formula for each of the following compounds. Use the table in question 1 to help you.

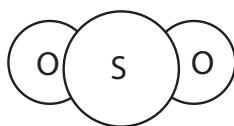
a magnesium carbonate	MgCO ₂	MgCO	MgCO ₃
b zinc sulfate	ZnSO ₃	ZnSO ₄	ZcSO ₄
c ammonium chloride	NH ₄ Cl	NH ₃ Cl	NH ₄ CL
d potassium nitrate	KNO ₂	KNO ₃	PNO ₃
e sodium hydrogen carbonate	SHCO ₃	NaHCO ₂	NaHCO ₃
f sodium hydroxide	SOH	NaOH	NaHO

Naming molecules

Molecules of a compound or element can be represented in diagrams. Below are two common molecules.



water



sulfur dioxide

1 Match the names and formula to the correct diagrams using the key below.

Key				
carbon	nitrogen	magnesium	oxygen	
zinc	sodium	sulfur	chlorine	potassium

Name	Formula	Molecular diagram
sodium nitrate	ZnCO_3	
magnesium sulphate	KCl	
potassium chloride	NaNO_3	
nitrogen dioxide	MgSO_4	
zinc carbonate	NO_2	

Objectives

- To distinguish between the properties and uses of acids and bases
- To identify different hazardous symbols and their meanings
- To understand that solutions can be acidic, basic or neutral and that we can use indicators to identify different solutions
- To understand that pH is a scale used to compare the acidity of different solutions

Prior knowledge

Students should understand that chemicals can be dangerous, and should be able to identify some common hazardous symbols seen in everyday life.

Background

Acids and bases are two important groups of chemical compounds, each with their own properties and uses. They can be detected using indicators such as litmus and universal indicator. Universal indicator can be used to find the pH of a substance. A substance's pH is a measure of its acidity, and it ranges from 1 to 14. Acidic substances have a pH below 7, neutral substances have a pH of 7 and basic substances have a pH above 7. Acids and bases can be dangerous so they must be handled carefully. Strong acids and bases are corrosive and must be correctly labelled.

Starter activity

Blindfold students and provide them with a piece of lemon, some natural yoghurt and a very small amount of baking soda to taste. Ask students, 'How would you describe the taste?' and 'What do you think each food is?' Students should use the words 'sour' and 'bitter' in their responses.

Activity sheets

The Activity sheet 'What are acids and bases?' introduces students to acids and bases. Students write a paragraph on the characteristics of acids using the notes provided and then add the missing vowels to complete a passage on bases. They are then asked to classify a number of different substances as either an acid or a base.

Before using acids and bases in the laboratory, students must be aware of the hazards associated with their use. The 'Hazardous symbols' Activity sheet asks students to identify different hazardous symbols and match each symbol to its meaning.

After completing the Activity sheet 'Indicators', students should be able to test a solution with litmus paper and identify it as an acid or a base. Students will need red and blue litmus paper, test tubes and a test-tube rack as well as solutions that are acidic, basic and neutral. Ensure students wear safety glasses (and other protective clothing as required) at all times.

The Activity sheet 'The pH scale' explains that pH is a measure of how acidic or basic a substance is. Students complete a cloze passage about the pH scale and then colour and label a diagram. They then choose the correct endings for the sentences provided.

'Acids and bases around the home' presents a crossword puzzle to familiarise students with different acids and bases found around the home. There are also some matching statements to complete.

Extension ideas

- Students could make and test a natural indicator made from red cabbage or grape juice.

Selected answers

Acids and bases around the home:

Across: 2 carbonic 6 bacteria 9 sodium bicarbonate 12 sulfuric **Down:** 1 sodium hydroxide 2 citric 3 cleaner 4 antacid 5 hazard 7 gloves 8 caustic 10 bases 11 corrosive



Work safely

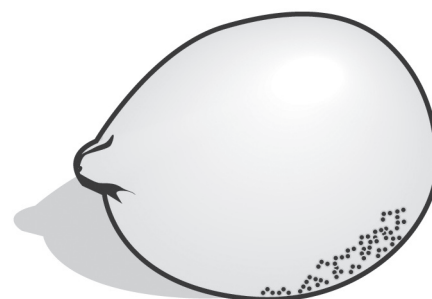
Teachers should conduct their own risk assessments for any activity or experiment they choose to do with their classes. All experiments should be carried out under close supervision.

What are acids and bases?

Acids and **bases** are two important groups of chemicals that have many different uses in everyday life.

 **1** Use the notes below to write a paragraph about **acids** in your workbook.

- very common substances used in everyday life
- contain the element hydrogen
- strong acids are dangerous because they are corrosive
- many foods are acidic, such as citrus fruits and vinegar
- a sour taste
- attack other substances
- common acids include citric acid, found in fruit, and acetic acid, which is vinegar
- common laboratory acids include hydrochloric, sulfuric and nitric acid



2 Fill in the missing vowels (a, e, i, o, u) to complete the passage below about bases.

Bases are the chem ___ c ___ l opposite of ac ___ ds. They have a b ___ tter t ___ ste and feel slippery or soapy to t ___ ch. They are often used around the h ___ me as cleaning agents. Like acids, some b ___ ses are very c ___ rr ___ sive and therefore can be very d ___ ng ___ ro ___ s to taste and touch. We say that those bases are c ___ ust ___ c. Strong bases, such as s ___ dium hydr ___ x ___ de and p ___ tassi ___ m hydroxide are found in ___ ven and dra ___ n cleaners. Other bases are used in s ___ ps, shampoos and d ___ shwashing liq ___ ids. Bases that d ___ ssolv ___ in w ___ t ___ r are called alkal ___ s.

3 Read the following list of substances. Underline the bases in red pen. Circle the acids in green.

- | | |
|--------------------|-----------------------|
| ● lemon juice | ● dishwashing liquid |
| ● sodium hydroxide | ● potassium hydroxide |
| ● vinegar | ● cola |
| ● drain cleaner | ● oven cleaner |
| ● nitric acid | ● lemonade |

Hazardous symbols

Strong acids and bases are very dangerous as they are **corrosive**. This means they can attack and destroy living tissue, such as eyes and skin. All dangerous chemicals have 'hazardous' symbols on them to warn people about the risks of using them.

 **1** Why is it important to have hazardous symbols on dangerous chemicals?

2 Draw a line between each symbol and the correct hazard.

irritant	
flammable	
toxic	
corrosive	
harmful	

3 Match the sentence halves below.

Corrosive substances	can make you ill, but are not as dangerous as toxic substances.
Irritants	destroy living tissues such as eyes and skin.
Harmful substances	can cause death if swallowed, breathed in or absorbed through the skin.
Flammable substances	are not corrosive but can make the skin go red or blister.
Toxic substances	catch on fire easily.

Indicators



Acids and **bases** are often colourless solutions, so it can be hard to tell them apart. To find out if a substance is an acid or a base we use **indicators**, which change colour when mixed with an acid or a base. **Litmus** is an indicator that is made from plants.

Experiment: What colour does litmus paper turn in acids and bases?

Aim: To find out the effect of acids and bases on litmus paper.

Equipment

- red and blue litmus paper
- 3 test tubes
- safety glasses
- labelled dropper bottles of acidic, basic and neutral substances
- test-tube rack

1 Use the words in the word bank to complete the method.

Method

- Put on safety glasses (and other protective clothing as advised by your teacher).
- Cut three 2-centimetre pieces each of red _____ paper and _____ litmus paper.
- Label each test _____ using a marker pen. Write 'acid' on one, '_____ ' on another and '_____ ' on the last one.
- Pour in 2 centimetres of each _____ into the correct _____ tube.
- Place a piece of _____ litmus paper in each test tube.
- Record any _____ change in the table below.
- Pour substances down the sink with running _____.
- Repeat the experiment, this time testing with blue litmus paper.

Word bank

neutral
substance
litmus
red
blue
water
tube
colour
test
base

2 Complete the experiment and record your results in the table below.

Type of substance	Effect on red litmus paper	Effect on blue litmus paper
acid		
base		
neutral		

The pH scale

Litmus paper can only tell you if a substance is an **acid** or a **base**. It cannot tell you how strong an acid or base it is. This is what tells you how dangerous it is. **Universal indicator** is a special indicator that can be used to measure the strength of an acid or base on the **pH scale**. The indicator changes colour depending on the strength of the acid or base.

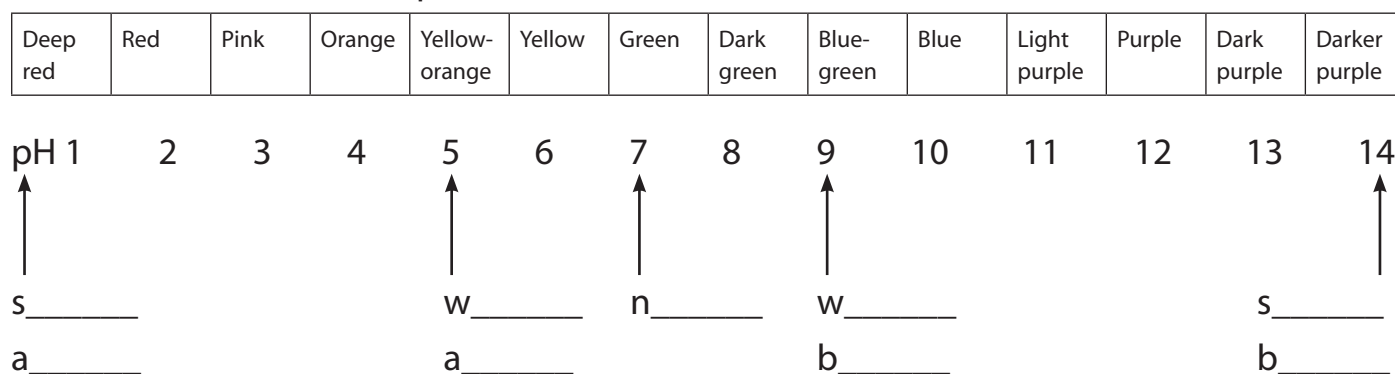
 **1** Use the words in the word bank to complete the passage.

You can describe how a _____ or how b _____ a substance is by using the numbers on the pH s _____. The pH scale ranges from 1 to 14. Numbers l _____ than pH 7 mean that substances are acidic. Numbers h _____ than pH 7 mean that the substance is basic. If a substance has a pH of 7, it is n _____ – neither acidic nor basic. Substances with a pH of 1 are s _____ acids and substances with a pH of _____ are strong bases.

Word bank

14
scale
acidic
neutral
basic
higher
lower
strong

2 Colour in and label the pH scale below.

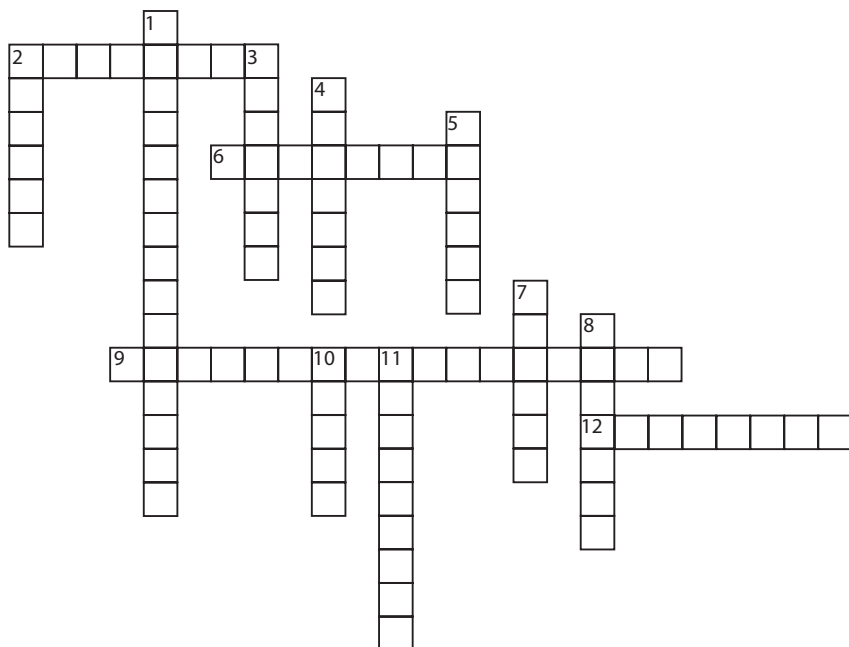


3 Circle the correct ending for each of the sentences below.

A substance that is pH 13 is	a strong acid.	a weak base.	a strong base.
A substance with a pH of 7 will turn	purple.	green.	red.
Acids have a pH	less than 7.	between 5 and 9.	between 8 and 14.

Acids and bases around the home

 **1** Complete the crossword using the words in the word bank.



Word bank

sodium
bicarbonate
antacid
hazard
sulfuric
corrosive
gloves
bacteria
caustic
sodium
hydroxide
cleaner
carbonic
citric
bases

Across

- 2** The bubbles in soft drink are formed by _____ acid.
- 6** Hydrochloric acid in your stomach kills _____.
- 9** A base used in cooking to help cakes rise
- 12** Car batteries contain _____ acid.

Down

- 1** An example of a strong base
- 2** A type of acid found in lemon juice
- 3** Ammonia is a common household _____.
- 4** These tablets are used to help settle upset stomachs.
- 5** _____ous symbols are placed on chemicals to warn us that they are dangerous.
- 7** You should always wear these when cleaning to keep your skin from becoming irritated.
- 8** Bases that destroy living tissue are sometimes called _____.
- 10** Most cleaning products around the home are _____.
- 11** Strong acids are dangerous because they are _____.

2 Match the statements below.

The acids we eat

citric acid when we eat fruit.

We eat acids such as

and they can attack your skin.

Bases are corrosive like acids

taste sour.

Bases are often found in cleaning products

like bleach and drain cleaner.

Objectives

- To identify reactants and products in chemical reactions
- To describe reactions using word equations
- To predict the products of different chemical reactions
- To investigate the way acids react with metals, bases and carbonates
- To understand how indigestion can be treated with the use of antacids

Prior knowledge

Students should have a basic understanding of, and be able to write, simple word equations.

Background

In a chemical reaction, new materials are formed. We use word equations to represent chemical reactions. The reactants are shown on the left and the products are on the right. When an acid reacts with a metal, hydrogen gas and a salt is formed. When an acid reacts with a carbonate, carbon dioxide gas is formed. A salt and water are also formed in the reaction. Neutralisation is a reaction between an acid and base that results in the formation of a salt and water.

Starter activity

Ask students if they have ever been stung by a jellyfish. What can you put on the sting to help it stop hurting? One suggestion should be vinegar. Ask students if vinegar is an acid or a base. Then ask them whether they think, if vinegar is an acid, that jellyfish venom is an acid or a base? Explain that acids and bases can be mixed to cancel each other out. Discuss the term 'neutralisation'. Finish by asking students, 'If you were stung by a bee that has acidic venom, would you put toothpaste or vinegar on the sting? Why?'

Activity sheets

After completing the Activity sheet 'How do we write chemical reactions?', students should be able to predict the reactants and products for different types of simple chemical reactions. Students are provided with the reactants and must determine the product that will result. Students are then provided with partially completed word equations and must

predict the missing reactant in each case. The activity concludes with some mixed word equations to solve.

The Activity sheet 'Reacting metals with acids' introduces the general word equation for reactions between a metal and an acid. Students choose the correct salt that forms from different acids, then read a passage and answer some questions to show their understanding of metal and acid reactions.

The 'Reacting acids with carbonates' Activity sheet explains how acids and carbonates react. Students choose the correct ending for different sentences. Students consolidate their understanding of reactions between acids and carbonates by predicting the name of the salt produced in a number of reactions.

'Neutralisation' explores reactions between acids and bases. Students answer questions about an experiment involving an acid and a base.

The Activity sheet 'Indigestion' explains a real-life application of neutralisation. This work sheet presents notes about indigestion. Students can expand the notes to write a paragraph on the topic before analysing the results of an experiment testing the effectiveness of different antacids.

Extension ideas

- Students could perform an experiment to find out which metals are the most reactive with sulfuric acid by measuring the temperature increase when each metal is combined with the acid.
- Students could make sherbet as an example of a neutralisation reaction.

Selected answers

Reacting acids with carbonates: 3 zinc chloride, copper nitrate, calcium nitrate, lead sulfate, copper sulfate

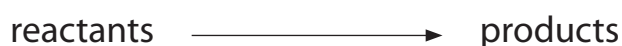


Work safely

Teachers should conduct their own risk assessments for any activity or experiment they choose to do with their classes. All experiments should be carried out under close supervision.

How do we write chemical reactions?

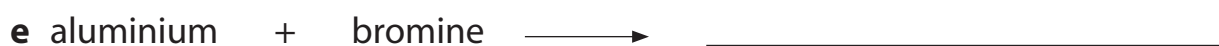
We use **word equations** to write down what happens in a **chemical reaction**. The equation shows the substances that are reacting (**reactants**) and what the reaction produces (**products**).



If you heat copper carbonate, a chemical reaction occurs and copper oxide and carbon dioxide are produced. This can be written as the following word equation:



 **1** Complete the following equations by writing in the correct **product** after the arrow. The first one has been done for you.



2 Complete the following equations by writing in the missing **reactant**.



3 Complete the following equations by filling in the missing **reactants** or **products**.



Reacting metals with acids

Acids are **corrosive** and eat some substances away. Some metals react when placed in acid. The piece of metal gets smaller, and bubbles of hydrogen gas are produced. The metal has not actually disappeared – it has changed into a new product, called a **salt**. As well as sodium chloride (common table salt), there are many other types of salts. The general **word equation** to show a reaction between an acid and a metal is:



The first part of the name of the salt that forms comes from the metal and the second part comes from the acid used. For example, if calcium reacts with hydrochloric acid, the salt that is produced is called calcium chloride.



1 Cross out the incorrect words from the options provided.

- a** Sulfuric acid forms chloride/nitrate/sulfate salts.
- b** Nitric acid forms chloride/nitrate/sulfate salts.
- c** Hydrochloric acid forms chloride/nitrate/sulfate salts.

2 Read the passage below and then answer the questions that follow.

Michael was conducting an experiment. When he placed magnesium in hydrochloric acid (hydrogen chloride), bubbles appeared, and the magnesium got smaller and disappeared. When he placed a lit match above the test tube, there was a squeaky pop. Michael knew that this meant the gas was hydrogen.

a What were the two reactants in Michael's experiment?

b Name one product.

c Look at the word equation below. It shows that there are two products produced. What do you think the missing product is?



Reacting acids with carbonates

Carbonates are substances that contain carbon and oxygen. When carbonates are mixed with **acids** they react. They fizz when combined because carbon dioxide is produced. Two other products are also produced: **a salt** and **water**. The general word equation for the reaction between a carbonate and an acid is:



We know that a salt is formed because if you evaporate the liquid that is left at the end of the reaction, a substance is left behind. The name of the salt that forms depends on the type of carbonate and the type of acid that react. The first part of the salt's name comes from the carbonate and the second part comes from the acid used.

 **1** Cross out the incorrect answers.

A reaction between a carbonate and an acid produces	hydrogen gas. carbon dioxide gas.
The first part of the salt's name comes from the	acid. carbonate.
The second part of the salt's name comes from the	acid. carbonate.

2 Name the salt produced when the following reactions occur. The first one has been done for you.

- a** magnesium carbonate + hydrochloric acid magnesium chloride
- b** zinc carbonate + hydrochloric acid _____
- c** copper carbonate + nitric acid _____
- d** calcium carbonate + nitric acid _____
- e** lead carbonate + sulfuric acid _____
- f** copper carbonate + sulfuric acid _____

Neutralisation

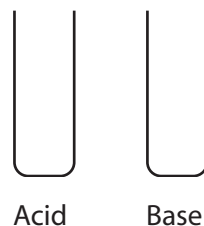
When an **acid** and a **base** are mixed, the chemical reaction that occurs is called **neutralisation**. The chemical reaction can be shown in the following word equation:



The name of the **salt** depends on the metal in the base used and on the acid used.



- 1 Sophia performed an experiment to find out how many drops of sodium hydroxide are needed to neutralise 5 mL of hydrochloric acid. Colour what each of these test tubes would look like if universal indicator was added.



- 2 What is the name of the base Sophia used in her experiment? _____
- 3 Sophia slowly added the base to the acid, one drop at a time, making sure she shook the test tube after each drop. Why do you think she needed to shake the test tube after each drop? _____
- 4 What colour would the solution be when neutralisation had occurred? _____
- 5 Sophia was not concentrating and put in a whole eye-dropper full of base at one time. The solution turned blue. What pH is the liquid now? _____
- 6 How could she get the solution back to neutral? _____
- 7 Use the word bank to complete the word equation for Sophia's experiment.

h _____ a _____ + s _____ h _____ $\longrightarrow$ s _____ c _____ + w _____

Word bank

sodium hydroxide sodium chloride hydrochloric acid water

Indigestion

- Indigestion occurs when your stomach produces too much acid and it backs up into your oesophagus.
- Only weak bases are safe enough to use in antacids.
- Your stomach contains hydrochloric acid.
- Indigestion tablets called antacids contain a base.
- Stomach acid helps digest your food and kill bacteria.
- The antacid tablets neutralise the extra acid.
- Some antacid tablets contain the bases calcium carbonate and magnesium hydroxide.

-  **1** Use the notes above to write a paragraph about indigestion in your workbook.
- 2** Sam and Alisha conducted an experiment to find out which was the most effective antacid. They added 5 drops of universal indicator to 20 mL of hydrochloric acid that was the same strength as stomach acid. They then added crushed antacid tablet one spatula at a time until the solution turned green. Their results are shown below.

Antacid tablet	Number of spatulas of antacid needed
A	5
B	7
C	8

- a** Why do you think they crushed the antacid tablets first? _____
- _____
- b** What would the pH of the solution be when it turned green? _____
- c** Which was the most effective antacid? _____
- d** How could Sam and Alisha more accurately measure the amount of antacid being added? _____
- _____

Objectives

- To understand the difference between endothermic and exothermic reactions
- To develop students' ability to plan and safely conduct experiments and to record and analyse data
- To identify the reactants and products formed during fermentation
- To use word equations to represent chemical equations
- To understand the difference between decomposition and synthesis reactions

Prior knowledge

Students should be able to identify reactants and products in simple chemical reactions and write word equations to represent chemical reactions.

Background

Chemical reactions either release energy into the environment or take in energy. Exothermic reactions release energy, usually in the form of heat, light or sound. Endothermic reactions take in energy, often in the form of heat. Cold packs involve an endothermic reaction – heat is taken in when chemicals are mixed. Chemistry also has important applications in the food industry. The process of fermentation is used in the production of wine, beer and bread. Chemical reactions can be classified as a number of different types. This unit explores two of the simpler types: decomposition and synthesis.

Starter activity

Ask students if they have ever had an injury while playing sport and if so, what first aid was applied. Were they ever given a cold pack? How was the pack made cold? How is a cold pack different to an ice pack? What do students think is inside a cold pack? Explain that cold packs use a chemical concept called endothermic reactions.

Activity sheets

The Activity sheet 'Energy changes in chemical reactions' explains the energy changes that occur during a chemical reaction. Students use the word bank to complete the passage about endothermic and exothermic reactions. They then use the information

to answer true/false questions before filling in the missing vowels to complete a table.

The Activity sheets 'Cold packs (1)' and 'Cold packs (2)' develop students' skills in safely conducting experiments and recording and analysing data, while investigating a real-life application of endothermic reactions used in cold packs. Students will need measuring scales, measuring cylinders, test tubes, thermometers, test-tube racks, safety glasses, stopwatches, distilled water, filter paper and rubber stoppers with holes. Each group will also need 2 grams of potassium nitrate, ammonium nitrate and ammonium chloride. Dispose of chemicals in accordance with your school's procedures.

In the Activity sheet 'Fermentation' students are introduced to fermentation. Students complete passages by filling in the missing vowels. There are also some short-answer questions to complete.

'Decomposition and synthesis' looks at two types of chemical reactions: decomposition and synthesis. Students complete a cloze passage before classifying diagrams and word equations as either decomposition or synthesis reactions.

Extension ideas

- Students could perform an experiment to identify exothermic and endothermic reactions.
- Students could investigate how variables such as sugar concentration and temperature influence the rate of fermentation in yeast.



Work safely

Teachers should conduct their own risk assessments for any activity or experiment they choose to do with their classes. All experiments should be carried out under close supervision.

Energy changes in chemical reactions

 **1** Use the words in the word bank to complete the passage below.

Whenever there is a chemical r_____ an energy
c_____ occurs. Some chemical reactions r_____ energy.
In other chemical reactions e_____ is needed to start the
reaction. Chemical reactions that release energy, normally in the form
of h_____, are **exothermic**. C_____ is an important
exothermic reaction. It releases heat energy, l_____ energy and
s_____ energy.

An e_____ reaction requires some form of energy to be
s_____ to start the reaction. **Endothermic** reactions often result
in a d_____ in temperature. Photosynthesis is an important
endothermic reaction. Plants use energy from the s_____ to
make food from c_____ dioxide and w_____.

Word bank

release
change
Combustion
supplied
decrease
water
sun
reaction
heat
endothermic
carbon
energy
light
sound

2 Decide whether each of the following statements is true or false.

- a** Exothermic reactions need energy supplied to start them. _____
- b** Exothermic reactions release energy, usually in the form of heat. _____
- c** Photosynthesis is an example of an exothermic reaction. _____
- d** Combustion is an example of an endothermic reaction. _____

3 Rewrite the false statements so they are true.

4 Fill in the missing vowels (a, e, i, o, u) to complete the table below.

	Energy is ...	Example
__nd __ th __ rm __ c r __ __ ct __ __ ns	__ bs __ rb __ d	ph __ t __ synth __ s __ s
__ x __ th __ rm __ c r __ __ ct __ __ ns	r __ l __ __ s __ d	c __ mb __ st __ __ n



Cold packs (1)

Cold packs work by using an **endothermic reaction**. The cold pack has two sealed bags: one containing water and the other containing a white powder called ammonium nitrate. The chemical reaction is activated by breaking the barrier separating the water and the ammonium nitrate. The water then mixes with the powder, dissolving it, and the solution becomes cold.

Experiment: Which chemical compound makes the best cold pack?

Aim: To see which chemical produces the biggest drop in temperature.

Equipment

- scales
- 3 thermometers
- measuring cylinder
- test-tube rack
- 3 test tubes
- distilled water
- 3 rubber stoppers with holes
- 3 pieces of filter paper
- stopwatch
- safety glasses
- chemicals: potassium nitrate, ammonium chloride, ammonium nitrate

1 Use the words in the word bank to complete the method.

Method

- Put on safety glasses (and any protective wear as advised by your teacher) and then weigh out 2 grams of each c _____ on filter paper.
- Label each test tube with the name of the chemical, then place in test-tube r _____.
- Measure out 10 mL of d _____ water and pour some into each test tube.
- Put the potassium nitrate in the t _____ t _____ labelled 'potassium nitrate', and start the stopwatch.
- Quickly put on r _____ stopper and record the temperature of the solution.
- Shake for 10 seconds.
- Record the t _____ every m _____ for 4 minutes.
- Repeat for the other two chemicals.

Word bank

test tube
temperature
chemical
rubber
minute
rack
distilled

Cold packs (2)



Complete the Activity sheet 'Cold packs (1)', then answer the following questions.

1 Identify one risk with this experiment. _____

2 State how you will address this risk. _____

3 Name two variables you need to control to make this a fair test.

a _____

b _____

4 Complete the experiment and record your results in the table below.

Chemical compound	Temperature (°C) after				
	0 min	1 min	2 min	3 min	4 min
potassium nitrate					
ammonium chloride					
ammonium nitrate					

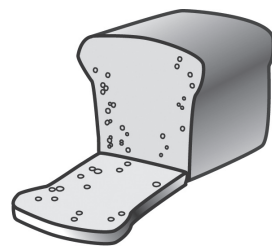
5 Draw a line graph of your results in your workbook. Draw one line for each chemical compound, using a different colour for each. Record the colours in a key.

6 Complete the conclusion for the experiment.

Conclusion: The chemical compound that had the biggest drop in temperature was _____, followed by _____ and _____.
The best chemical compound for cold packs would be _____.

Fermentation

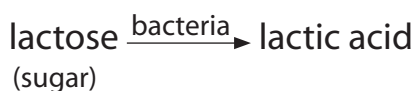
For hundreds of years people have used microorganisms such as yeast and bacteria to make foods and drinks such as beer, yoghurt, wine and cheese using a process called **fermentation**. Fermentation is the process where an organism converts a carbohydrate (starch or sugar) into alcohol or an acid.



 **1** Fill in the missing vowels (a, e, i, o, u) to complete the text below.

Bacteria

B _ _ ct _ _ ri _ _ are m _ _ cr _ _ _ rgan _ _ sms used in the process of f _ _ rm _ _ nt _ _ ti _ _ n to make yoghurt and ch _ _ _ se. The b _ _ ct _ _ r _ _ a feed on lactose, which is a s _ _ g _ _ r in milk, converting it to lactic _ _ c _ _ d. The increased acidity thickens and s _ _ urs the m _ _ lk. The word equation for the fermentation of milk is:



Yeast

Y _ _ _ st is a single-celled microscopic f _ _ ng _ _ s that uses s _ _ g _ _ r as food. Yeast is used to make br _ _ _ d, b _ _ _ r and w _ _ n _ _ . When bread is made, the y _ _ _ st feeds on sugar (gl _ _ c _ _ s _ _) in the bread d _ _ _ gh. The yeast is used to make b _ _ bbl _ _ s in the dough. As the dough starts to f _ _ rm _ _ nt, c _ _ rb _ _ n d _ _ _ x _ _ d _ _ is produced. The carbon dioxide is tr _ _ pp _ _ d in the dough, causing it to r _ _ s _ _ . The w _ _ rd _ _ qu _ _ t _ _ on for this is:



When the dough is baked, the yeast is k _ _ ll _ _ d and the alcohol _ _ v _ _ por _ _ t _ _ s. In the br _ _ w _ _ ng of beer, the alc _ _ h _ _ l remains, along with the carbon dioxide, which gives b _ _ _ r its fizz.

2 What is the reactant in the process of fermentation used to make bread? _____

3 What are the products in the process of fermentation used to make bread? _____

4 What product do bacteria produce during the process of fermentation? _____

Decomposition and synthesis

There are many different types of chemical reactions. **Decomposition** and **synthesis** are just two. 'Decompose' means to break down, and 'synthesise' means to combine.

Reaction	Definition	Equation
decomposition reaction	A reaction where one compound breaks down into two or more simpler substances	$AB \longrightarrow A + B$
synthesis reaction	A reaction where two or more substances combine to form a more complex substance	$A + B \longrightarrow AB$

 **1** Use the words in the word bank to complete the following sentences.

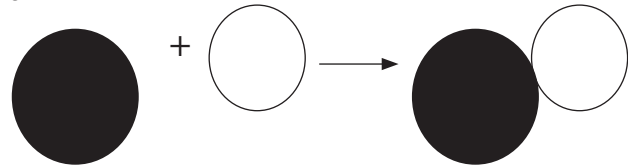
- Decomposition reactions require e_____, either in the form of h_____, light or e_____.
- The a_____ energy causes the reacting substance to b_____ down.
- Synthesis reactions are e_____.
- C_____ decompose but e_____ cannot.

Word bank

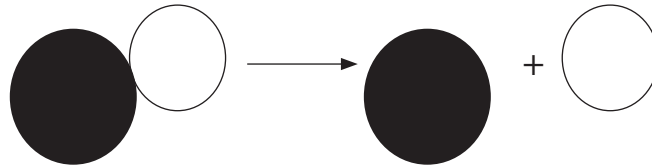
absorbed
compounds
energy
elements
heat
break
endothermic
electricity

2 Which type of reaction does each diagram represent?

a



b



3 Below are some different word equations. Decide whether each shows a synthesis or decomposition reaction.

a magnesium + oxygen $\xrightarrow{\text{heat}}$ magnesium oxide _____

b calcium carbonate $\xrightarrow{\text{electricity}}$ calcium oxide + carbon dioxide _____

c water _____ $\rightarrow$ hydrogen gas + oxygen gas _____

Objectives

- To understand that different chemical reactions occur at different rates
- To understand that the rate of reaction depends on concentration, surface area and temperature
- To identify different ways that chemical reactions can be sped up or slowed down
- To understand the role of energy in chemical reactions

Prior knowledge

Students will need to be able to create a line graph and a scientific drawing. Students should also have an understanding of variables and fair testing.

Background

Chemical reactions take place at different speeds. This is known as the rate of reaction. Reactions happen when particles of the reactants collide with each other. We can increase or decrease the rate of reaction by increasing or decreasing the number of collisions and the energy transferred during the collisions. Temperature, concentration, surface area and catalysts all affect reaction rates.

Starter activity

Show students the way that ashes catalyse the burning of sugar. A sugar cube will not burn unless it is first rubbed with ash. To demonstrate this, first hold a sugar cube in the flame of a Bunsen burner. It will melt, but will not burn. Then rub a sugar cube with ash before holding it in the flame – it will now burn.

Resource and Activity sheets

The Resource sheet 'Changing the rate of reaction (1)' and the Activity sheet 'Changing the rate of reaction (2)' develop students' understanding of the different factors that affect the rate of reaction. Students use the information provided in the Resource sheet to complete a cloze passage then complete sentences and answer true/false questions.

The Activity sheets 'Temperature and reaction rates (1)' and (2)' develop students' skills in safely conducting experiments and recording and analysing data, while investigating how temperature affects reaction rates.

Cold acid should be stored in the fridge. For warm acid, place sufficient acid for the class in a water bath set at 45°C. Use 1M or 2M hydrochloric acid.

The Activity sheet 'Surface area and reaction rates' presents results from an experiment investigating the effect of surface area on the reaction rate between magnesium and hydrochloric acid. Students write the word equation for the reaction, then graph and analyse the results.

Extension ideas

- Students could perform an experiment to investigate the effect of the concentration of hydrochloric acid on the rate of its reaction with magnesium ribbon.
- Students could investigate the effect of enzymes in detergents and cleaning products.
- Students could investigate the effect of different factors on the rate of reaction of acid on marble.



Work safely

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Changing the rate of reaction (1)

Chemical reactions take place at different speeds. Some reactions, such as explosions, take place very quickly, while others, like the rusting of a car, can take years. The rate of a reaction is a measure of how quickly a chemical reaction occurs.



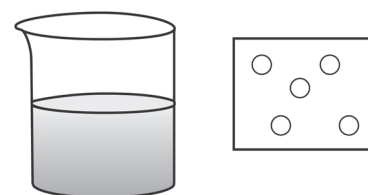
Reactions take place when particles collide with a certain amount of energy. Increasing the number of collisions and the energy that the particles collide with increases the rate of reaction. You can speed up or slow down a reaction by changing a number of different conditions.

1 Temperature

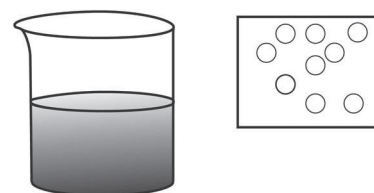
Increasing temperature speeds up the rate of reaction. As the temperature increases, the particles move faster and collide more often. This is why food is placed in fridges to slow down the chemical reactions that cause food to go off.

2 Concentration of solution

The concentration of a solution refers to the amount of a substance that is present in a certain amount of water. The more concentrated a solution is, the faster the reaction will occur, because there are more particles to collide into each other.



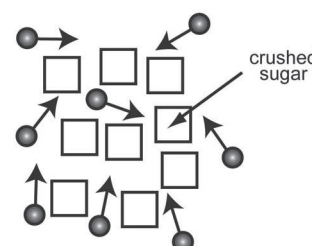
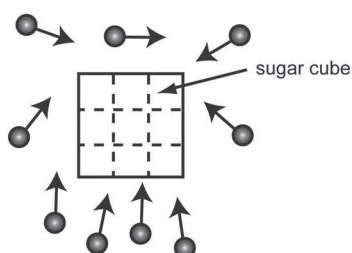
Less concentrated solution



More concentrated solution

3 Size of particles

When a solid is crushed into smaller pieces there is more surface area in contact with the liquid. This means the particles collide with the liquid more often, speeding up the reaction.



4 Catalysts

Catalysts are substances that speed up the rate of reaction without being used up in the reaction.

Changing the rate of reaction (2)

Read the Resource sheet 'Changing the rate of reaction (1)', then answer the following questions.

 **1** Use the words in the word bank to complete the passage.

Reactant p _____ must collide for a r _____ between them to take place. The r _____ of reaction increases when different reactant particles c _____ into each other more often and with more e _____. The reaction rate can be sped up by increasing the s _____ a _____ of a solid, increasing the c _____ of a solution or increasing the t _____. Chemicals called c _____ can also be used to speed up c _____ reactions.

Word bank

collide
surface area
catalysts
particles
rate
concentration
chemical
reaction
energy
temperature

2 Cross out the incorrect item from each pair in brackets.

As the temperature increases, the particles move (faster/slower) and collide (more/less) often.

The more concentrated a solution, the (faster/slower) the reaction rate.

When a solid is (crushed/whole), there is a bigger surface area for reactions to take place.

Catalysts are substances that (speed up/slow down) the rate of reaction and are (used up/not used up) during the reaction.

3 Are these statements true or false?

- a** Explosions are reactions that occur slowly. _____
- b** The rate of reaction is a measure of how quickly a chemical reaction occurs. _____
- c** Increasing the number of times and the energy with which reactant particles collide decreases the rate of reaction. _____

4 Rewrite the false statements to make them true.

Temperature and reaction rates (1)



Temperature is one condition that can affect reaction rate. Most reactions increase their rate when the temperature is increased.

Experiment: The effect of temperature on the rate of reaction between magnesium ribbon and hydrochloric acid

Hypothesis: Increasing the temperature will speed up the rate of reaction between magnesium ribbon and hydrochloric acid.

Equipment

- 3 test tubes
- 3 x 2-cm strips of magnesium ribbon
- test-tube rack
- measuring cylinder
- hydrochloric acid at 3 different temperatures: cold, room temperature, warm
- stopwatch
- safety glasses and gloves

1 Number the steps of the method in the correct order.

Method

- _____ Measure temperature of acid and record in results table.
- _____ Collect equipment.
- _____ Measure out 10 mL of cold hydrochloric acid and pour into test tube.
- _____ Put on safety glasses and gloves.
- _____ Record reaction time in table.
- _____ Add a strip of magnesium and start stopwatch.
- _____ Repeat for room temperature and warm hydrochloric acid.
- _____ Stop timer when all the magnesium has reacted.

2 Hydrochloric acid has the hazardous symbol shown on its bottle. What does this mean? _____



3 Identify one risk with this experiment.

Temperature and reaction rates (2)



Complete the Activity sheet 'Temperature and reaction rates (1)', then answer the following questions.

1 What is the independent variable in the experiment? _____

2 What is the dependent variable in the experiment? _____

3 Name two variables that must be controlled to make it a fair test.

4 In your workbook, draw a scientific diagram of your experiment. When drawing a scientific diagram, remember to:

- use a ruler and pencil
- draw a 2D cross-section
- include labels.

5 Complete the experiment and record your results in the table.

	Cold acid	Room temp. acid	Warm acid
Temperature (°C)			
Reaction times			

6 Draw a bar graph of your results in your workbook. Remember to include a key.

7 Which reaction had the fastest rate? _____

8 Why do you think this was? _____

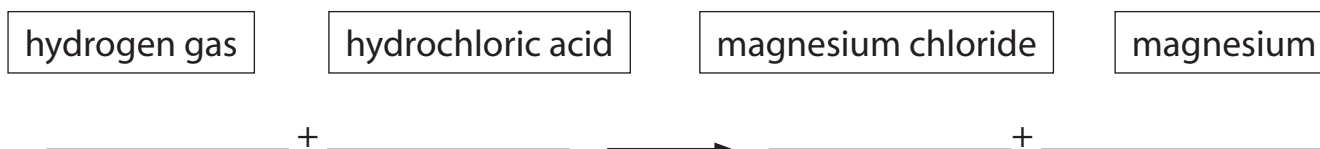
9 Complete the conclusion.

Conclusion: The _____ the temperature of hydrochloric acid, the _____ it reacted with the magnesium ribbon.

Surface area and reaction rates

Shelby conducted an experiment to investigate the effect of surface area on the reaction rate between hydrochloric acid and magnesium. She compared powdered magnesium to a piece of magnesium with the same mass. She placed the conical flask containing the magnesium and acid on electronic scales and recorded the loss of mass over a period of time. When there was no longer a loss in mass, she knew the reaction had finished.

-  **1** Listed below are the two reactants and two products for this experiment. Decide which are which, and then fill in the word equation.



- 2** Which product do you think was able to escape the conical flask? _____

Shelby's results are recorded below.

	0 min	1 min	2 min	3 min	4 min	5 min	6 min	7min	8 min	9 min
powdered magnesium (loss of mass, g)	0	0.4	0.7	0.8	0.85	0.9	0.9	0.9	0.9	0.9
piece of magnesium	0	0.25	0.3	0.45	0.6	0.7	0.75	0.85	0.9	0.9

- 3** Draw a line graph of her results in your workbook. Draw one line for powdered magnesium and one line for the piece of magnesium. Include a key.
- 4** Which type of magnesium has the fastest rate of reaction? _____
- 5** When did the reaction finish for the:
- a** powder? _____
- b** piece of magnesium? _____
- 6** Why do you think this happened? _____
- _____

Objectives

- To describe the characteristics of limestone
- To explain how chemistry can be used to produce a range of useful substances
- To understand that crude oil can be refined to produce a number of useful products
- To be aware of different fertilisers and the problems associated with their use
- To know that chemical reactions are involved in cooking

Prior knowledge

Students should have an understanding of chemical formulas and molecular diagrams.

Background

Scientists are able to produce a range of materials with specific properties to meet specific needs. Chemistry is used to refine crude oil and turn it into useful products such as kerosene, petrol, diesel and lubricating oils. Refining crude oil involves separating and purifying the different fractions and converting them to useful products. Simple ingredients such as baking soda, used in baking cakes, involve chemistry. Agriculture is another area that relies on chemistry, which is used to produce fertilisers that increase yield.

Starter activity

Introduce the topic of chemical resources by asking students to name some important chemical resources that we rely on in everyday life. Suggestions should include examples such as petrol and oil. Ask students to name some fossil fuels. Why are they called fossil fuels? What do we use fossil fuels for? What are some products that we make from oil? What problems are associated with the use of fossil fuels? How are we trying to reduce our reliance on fossil fuels? Discuss renewable and non-renewable energy sources.

Activity and Resource sheets

The Resource sheet 'Refining crude oil' and the Activity sheet 'Crude oil' explain the process of refining crude oil to obtain useful substances that we use in everyday life. Students use the Resource sheet to help them answer some short-answer questions. There are also some sentence endings about crude oil and some key words to complete.

In 'Limestone', students read a passage describing limestone, then underline characteristics of limestone before writing the characteristics in point form. There is also a sentence to complete before students use the key provided to draw a molecular diagram of calcium carbonate (limestone). Students match some phrases before choosing the correct ending for sentences.

The Activity sheet 'Fertilisers' describes the importance of fertilisers in agriculture as well as the problems associated with their use. Students identify correctly spelled words in a passage before identifying the essential elements in different fertilisers. Students are also asked to order steps to show the process of eutrophication.

The Activity sheet 'Chemistry in cooking' introduces the common chemical compound baking soda and explores its applications in cooking. Students read a passage before answering questions and drawing a molecular diagram. Students complete the activity by identifying the products formed when baking soda decomposes before writing a word equation.

Extension ideas

- Students could perform an experiment to investigate the effect of acid on limestone and marble.
- Students could investigate the effect of fertiliser on plant growth by growing cereals and exploring different rates of fertiliser application and the effects of this.
- Students could complete a cooking activity investigating the effect of different leaving agents on the quality of cupcakes.



Work safely

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Refining crude oil

Crude oil is a **fossil fuel** that formed millions of years ago from buried dead plant and animal matter. These remains were turned into fossil fuels by **heat** and **pressure**. Crude oil is a mixture of different compounds. Most of the compounds in crude oil are **hydrocarbons**. The only atoms they contain are carbon and hydrogen.

Some common hydrocarbons found in crude oil	Chemical formula
methane	CH_4
ethane	C_2H_6
propane	C_3H_{10}
butane	C_4H_{10}

Crude oil is separated into a number of different parts, called fractions, using a process called **fractional distillation**. This involves heating the crude oil until it **evaporates** and then letting the gas **condense**. The different hydrocarbons can be separated because they have different boiling points. The fractions are then processed into more useful hydrocarbons. Crude oil provides useful substances such as petrol, waxes, plastics, detergents, bitumen and man-made fibres such as nylon.

The diagram shows the different substances formed from fractional distillation. Substances with a high boiling point condense at the bottom and substances with lower boiling points condense at the top.

	Fraction	Boiling point (°C)
	LPG (liquefied petroleum gas) (used for heating and cooking)	<25
	petrol (fuel used for cars)	25–60
	naphtha (used for making chemicals)	60–180
	paraffin (flammable waxy solid used in candles and polishes)	180–220
	diesel (liquid fuel similar to petrol, used in cars, trucks and heavy machinery)	220–250
	lubricating oil (thick oil used to reduce friction between moving parts in machinery)	300–350
	bitumen (thick black substance used on roads)	>350

Fractional distillation of crude oil

Crude oil

Use the Resource sheet 'Refining crude oil' to answer these questions.



1 Cross out any answers that are incorrect.

Crude oil formed	from dead plants and animals millions of years ago thousands of years ago
Crude oil contains the elements	carbon oxygen hydrogen
The remains of living things are changed into fossil fuels by the effects of	air pressure heat
Most of the compounds in crude oil are	metals hydrocarbons

2 Fill in the missing letters in these important words from the Resource sheet.

c _ ud _ o _ l hy _ r _ ca _ b _ _ s d _ st _ _ _ ati _ _ b _ _ l _ _ g po _ _ ts

3 List four useful substances that can be produced from crude oil.

- _____
- _____
- _____
- _____

4 Use the diagram on the Resource sheet to answer these questions.

a The different fractions in crude oil are separated based on what characteristic?

b What is the boiling point of the following fractions?

lubricating oil _____ paraffin _____ petrol _____

c Which fraction has the lowest boiling point? _____

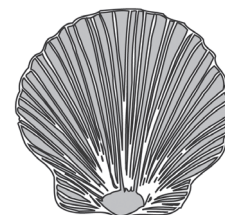
d Where does it condense in the tower? _____

e What type of fraction is used for fuel in trucks? _____

Limestone

1 Read the passage below and underline five characteristics of limestone.

Limestone is a type of rock that formed millions of years ago from the shells of dead ocean creatures. The shells built up on the sea floor and, over time, changed into rock. The chemical name for limestone is **calcium carbonate**. It has the chemical formula CaCO_3 . Limestone is dug out of the ground and used for many things, especially as a building material. It is used for making cement, mortar, concrete and building blocks.



Limestone can be damaged by **acid rain**. When fossil fuels are burnt they release chemical pollutants such as sulfur dioxide and nitrogen dioxide. These chemicals react with water in the atmosphere to form acid rain. Normally rain is slightly acidic, but anything with a pH of less than 5.6 is known as acid rain. Acid rain eats away at limestone, causing it to crumble.

2 In your workbook, write down the five characteristics you underlined in point form.

3 Complete the following sentence.

A molecule of calcium carbonate or CaCO_3 is made up of one atom of _____ and _____, and _____ atoms of oxygen.

4 Using the key, draw a molecule of calcium carbonate.

5 Match the sentence halves to complete the statements below.

When fossil fuels are burnt they

with water to form acidic rain.

The chemical pollutants released combine

to form sulfuric acid.

Acidic rain has a pH of less

release sulfur dioxide and nitrogen dioxide.

Sulfur dioxide combines with water

than 5.6.

Key	
	Calcium
	Carbon
	Oxygen

6 Choose the correct ending for each of the statements below.

The chemical formula for limestone is	CaCO_2 .	CaCO_3 .
The chemical formula for nitrogen dioxide is	NO_2 .	NO_3 .
The chemical formula for sulfuric acid is	HNO_3 .	H_2SO_4 .

Fertilisers

 1 Circle the correctly spelled word in each pair.

Fertilisers are chemicals/chemcials used to increase crop yield by putting essential eliments/elements such as phosphorus and potassium/potasium back into the soil. They also provide nitrogen, which increases plant growth. Fertilyzers/Fertilisers dissolve in water and are then absorbed by the plant. Fertilisers are salts that are made by neutralising different acids and bases.

The problem with fertilisers is that they can pollute/polute water supplies and cause the growth of algae/algee in waterways. This leads to a process called **eutrophication**. Lakes and rivers become 'dead' due to pollution from fertilisers.

2 What is the chemical symbol for each of the following essential elements?

nitrogen _____ phosphorus _____ potassium _____

3 The table below lists some common fertilisers. Identify the essential elements in each fertiliser that help increase crop yield. The first one has been done for you.

ammonium nitrate	$\text{NH}_4 \text{NO}_3$	nitrogen
ammonium sulfate	$(\text{NH}_4)_2\text{SO}_4$	
ammonium phosphate	$(\text{NH}_4)_3\text{PO}_4$	
potassium nitrate	KNO_3	

4 Number the steps below to show the order they occur in process of eutrophication.

_____ Other plants die.

_____ Fertilisers run off into the waterways.

_____ Algal bloom appears.

_____ Most organisms in the water die.

_____ Increase in amount of nitrate or phosphate in the water.

_____ Extra algae blocks off light to other plants.

_____ Bacteria feed on dead plants and multiply, using up oxygen in the water.

Chemistry in cooking

Baking soda is a chemical compound used when cooking cakes. The chemical name for baking soda is **sodium hydrogen carbonate**. It has the chemical formula NaHCO_3 . Baking soda is used in baking because when it is heated, it breaks down to form sodium carbonate, water and carbon dioxide gas. The carbon dioxide gas expands, making the cake rise. The trapped carbon dioxide gas makes the cake light and fluffy.

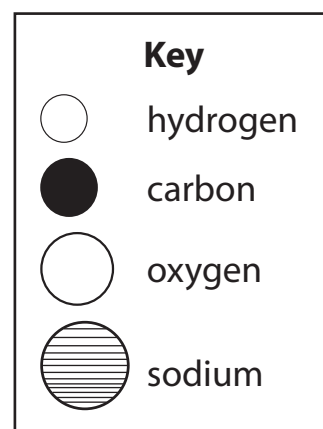
 1 Complete the table below.

Common name	Chemical name	Chemical formula	Elements present
Baking soda			

2 Match each element to the number of atoms present in a molecule of baking soda.

sodium	carbon	hydrogen	oxygen
1	1	3	1

3 Using the key provided, draw a diagram of a molecule of baking soda.



4 Heating baking soda is a **decomposition** reaction. This means that one chemical breaks down into two or more simpler substances. How many substances does baking soda break down into when heated? _____

5 Name these substances. _____

6 Complete the word equation for the decomposition of baking soda (sodium hydrogen carbonate).

_____ $\xrightarrow{\text{heat}}$ _____ + _____ + _____

Objectives

- To use word and symbol equations to represent chemical reactions
- To model chemical reactions in terms of rearrangement of atoms
- To understand that the mass of the reactants is equal to the mass of the products
- To understand that the conservation of mass in a chemical reaction can be demonstrated by simple chemical equations
- To balance simple chemical equations
- To develop students' ability to plan and safely conduct experiments and to record and analyse data

Prior knowledge

Students should be able to identify the reactants and products in a chemical equation. Students should also know the symbols for common elements and the formulas for common compounds.

Background

During chemical reactions, no matter is created, lost or destroyed. This means that mass is conserved. The mass of the reactants is equal to the mass of the products. The total mass remains unchanged – the atoms in the reactants are merely rearranged to form new substances or products. When equations are written to represent chemical reactions, they must be balanced. The total number of atoms of each element on the left must always equal the total number on the right. We balance chemical equations by placing a large number called a coefficient in front of the formula. This multiplies the whole of the formula.

Starter activity

Demonstrate the chemical reaction between acetic acid and sodium hydrogen carbonate. Measure 500 mL of acetic acid into a 2L plastic bottle and weigh on digital scales. Explain that when sodium bicarbonate and acetic acid are mixed, an immediate reaction occurs, producing a gas. Ask students what they think will happen to the mass of the bottle and its contents when 5 g of sodium bicarbonate is added to the bottle and a gas is produced. Complete the demonstration, ensuring that the lid is placed on the bottle. Reweigh the bottle after the reaction has

finished. Discuss the results and then ask students, 'Would the result be the same if the lid was not on the bottle?'. Discuss this as a class and then repeat the demonstration, this time without the lid. Ask students why the mass changed this time.

Resource and Activity sheets

The Resource and Activity sheets 'Conserving mass (1) and (2)' explain how mass is conserved in chemical reactions. The Resource sheet demonstrates this scientific principle using different examples of chemical reactions. Students should refer to this page when completing the Activity sheet.

The Activity sheets 'Burning magnesium (1) and (2)' develop students' skills in safely conducting an experiment. Cut the magnesium ribbon into 5-cm lengths before the lesson, and ensure that students do not look directly at the flame, to avoid damaging their eyes.

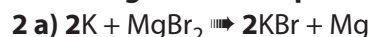
The Activity sheet 'Balancing chemical equations' shows students how to balance equations using trial and error, without drawing molecular diagrams. Students are provided with an example before they practise counting the number of atoms present when there are multiple numbers of a molecule. Finally, students try balancing equations.

Extension idea

- Students could practise balancing equations using an online simulation or game. Suggested search tag is 'balance chemical equations game'.

Selected answers

Balancing chemical equations:

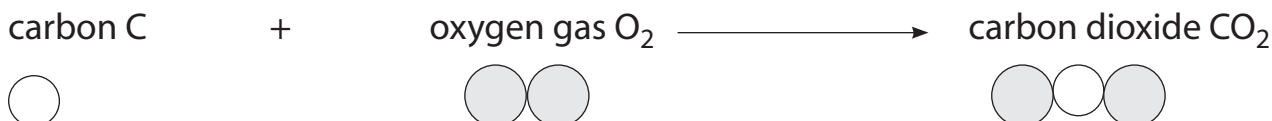


Work safely

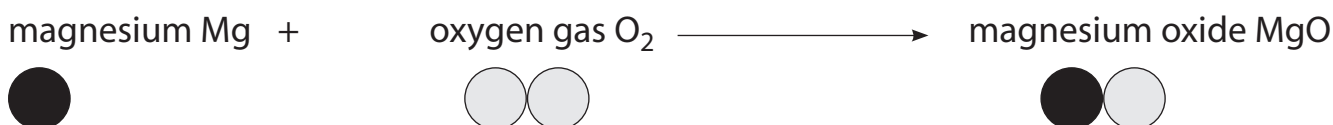
Teachers should conduct their own risk assessments for any activity or experiment they choose to do with their classes. All experiments should be carried out under close supervision.

Conserving mass (1)

When a chemical reaction occurs, the atoms in the reactants are rearranged to form the products. No new atoms are created, and none are destroyed. Mass is **conserved**. This means that the mass of the products at the end of the reaction is equal to the mass of the reactants at the beginning. We use **molecular drawings** to show this.



The number of atoms of each element at the start is the same as at the end, so we say that the equation is balanced. What if the equation is not balanced?



Something is wrong. There are more oxygen atoms at the start of the reaction than at the end, but we know that atoms cannot be lost during a chemical reaction. If there were two molecules of magnesium oxide at the end we would have the correct number of oxygen atoms, but one too many magnesium atoms. There must have been 2 magnesium atoms at the start of the reaction.

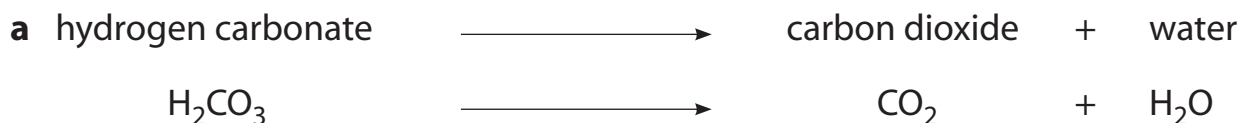


We say that this equation is balanced. We place a large number at the front to show how many molecules of each reactant and product there are. Placing a large number at the front of a formula multiplies the whole of the formula by that number. For example, **2MgO** means 2 atoms of magnesium **and** 2 atoms of oxygen.

Conserving mass (2)

Read the Resource sheet 'Conserving mass (1)' then answer the following questions.

-  **1** Below are two different chemical equations. Draw molecular diagrams to show how the atoms are rearranged and then decide if they are balanced or not. Remember to use different colours for different types of atoms.



At the start there are

_____ hydrogen atoms

_____ carbon atoms

_____ oxygen atoms

Total number of atoms =

At the end there are

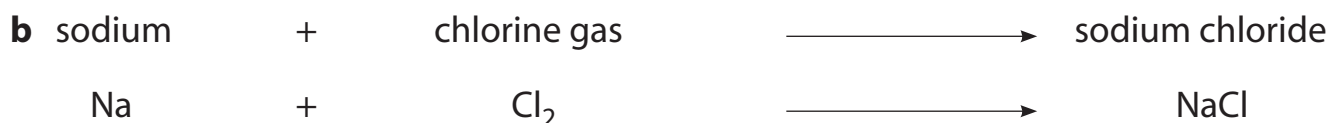
_____ hydrogen atoms

_____ carbon atoms

_____ oxygen atoms

Total number of atoms =

Is the equation balanced? If not, try increasing the number of different molecules. Remember to put a number at the front to show how many are needed.



At the start there are

_____ sodium atoms

_____ chlorine atoms

Total number of atoms =

At the end there are

_____ sodium atoms

_____ chlorine atoms

Total number of atoms =

Is the equation balanced? If not, try increasing the number of different molecules. Remember to put a number at the front to show how many are needed.

Burning magnesium (1)



When metal burns in air, a metal oxide is formed. The general word equation for the burning of metal in air is: metal + oxygen $\longrightarrow$ metal oxide.

Experiment: Burning magnesium and conservation of mass

Aim: To investigate what happens when magnesium burns.

1 Fill in the missing vowels (a, e, i, o, u) to complete the list of equipment.

Equipment

- cr _ c _ ble with l _ d
- 5-cm piece of m _ gn _ sium
- tr _ p _ d
- B _ ns _ n b _ rn _ r
- st _ _ l wool
- _ lectr _ n _ c sc _ les
- p _ pe cl _ y triangle
- gloves
- m _ t _ l tongs
- s _ fety glasses

2 Use the words in the word bank to complete the method.

Method

- Weigh the c _ _ _ _ _ and lid and record its mass.
- Put on g _ _ _ _ _ and safety glasses.
- Clean the magnesium with the steel w _ _ _ _ _ . Fold up the magnesium so it fits in the crucible. Reweigh the crucible and lid with the m _ _ _ _ _ inside and record this figure in the t _ _ _ _ _ .
- Set up bunsen burner, tripod and pipe clay triangle. Place crucible on top and heat. Do not let the white s _ _ _ _ _ escape.
- If the magnesium is not burning, it needs more o _ _ _ _ _ . If this is the case, carefully lift the lid slightly with the m _ _ _ _ _ tongs. The magnesium should b _ _ _ _ _ into flames. Replace the lid. **SAFETY PRECAUTION: DO NOT LOOK DIRECTLY AT FLAME.**
- After one m _ _ _ _ _ , lift lid again and let some a _ _ _ _ _ in. Repeat until the magnesium doesn't flare up again.
- Once the crucible has cooled down, reweigh it.

Word bank

magnesium
crucible
wool
table
oxygen
minute
burst
gloves
metal
smoke
air



Burning magnesium (2)

Complete the Activity sheet 'Burning magnesium (1)', then answer the following questions.

- 1 Identify one risk with this experiment. _____
- 2 How will you address this? _____
- 3 What is the name of the burnt product? _____
- 4 Complete the experiment and record your results in the table below.

	Mass (grams)
Mass of empty crucible and lid (A)	
Mass of crucible, lid and strip of magnesium (B)	
Mass of magnesium (B – A)	
Mass of magnesium oxide, crucible and lid (C)	
Mass of magnesium oxide (C – A)	

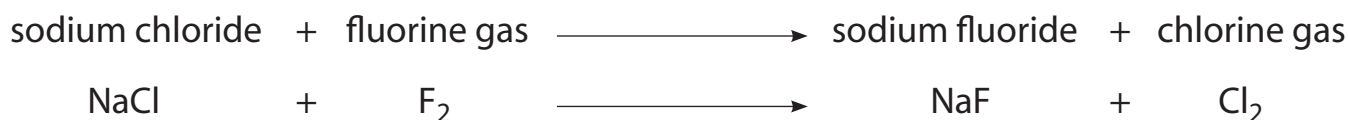
- 5 Write a word equation for the burning of magnesium.
 _____ + _____ → _____
- 6 What was the mass of the magnesium at the start? _____
- 7 What was the mass of the magnesium oxide at the end? _____
- 8 Was there any change in mass? Has the mass increased or decreased? _____
- 9 Why is the mass of the magnesium oxide greater than the mass of the magnesium?

- 10 We know that the mass of the reactants must be the same as the mass of the products.
 We can work out how much oxygen reacted with the magnesium because mass is conserved. Complete the summary below.

In this experiment, _____ g of magnesium produced _____ g of magnesium oxide. If we take the mass of the _____ from the mass of the magnesium oxide, we get the m_____ of the oxygen used in the reaction. This experiment used _____ g of oxygen.

Balancing chemical equations

You do not need to draw molecular diagrams each time to balance chemical equations. Look at the example below.



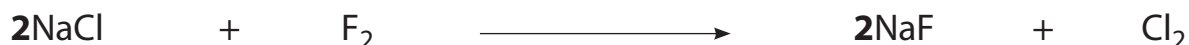
At the start there is

1 atom Na + 1 atom Cl + 2 atoms F = 4 atoms in total

At the end there is

1 atom Na + 2 atoms Cl + 1 atom F = 4 atoms in total

Even though the total number of atoms is the same, the equation is not balanced because we have more atoms of fluorine and less of chlorine at the start of the reaction. We use trial and error to see what numbers we need to add to the front of the compounds.



If we put a 2 in front of the NaCl we now have 2 atoms of chlorine each side but now we have 2 atoms of sodium at the start of the reaction, so we also have to put a two at the front of NaF. Now there are two fluorine atoms on each side, so the equation is balanced.

Remember: A large number at the front means you multiply the number of atoms in one molecule by that number.

 **1** Count the number of atoms in the following. The first one has been done.

a $2\text{Na}_2\text{SO}_4$ 4 atoms of sodium, 2 atoms of sulfur, 8 atoms of oxygen

b 3CH_4 _____

c $2\text{H}_2\text{O}$ _____

d 2CaCO_3 _____

2 Now try balancing the following equations in your workbook.

a $__\text{K} + __\text{MgBr}_2 \longrightarrow __\text{KBr} + __\text{Mg}$

b $__\text{H}_2\text{O}_2 \longrightarrow __\text{H}_2\text{O} + __\text{O}_2$

c $__\text{NO}_2 \longrightarrow __\text{O}_2 + __\text{N}_2$

d $__\text{N}_2 + __\text{H}_2 \longrightarrow __\text{NH}_3$

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